

Calculations of complex chemical reaction equilibria using stoichiometric and non-stoichiometric approaches in combination with arc-length continuation

Housam Binous¹  | Ahmed Bellagi² 

¹Chemical Engineering Department, National Institute of Applied Sciences & Technology, University of Carthage, Tunis, Tunisia

²Energy Engineering Department, Ecole Nationale d'Ingénieurs de Monastir - ENIM, University of Monastir, Monastir, Tunisia

Correspondence

Housam Binous, Chemical Engineering Department, National Institute of Applied Sciences & Technology, University of Carthage, Tunis, Tunisia.

Email: binoushousam@yahoo.com

Abstract

Chemical equilibrium calculations—as they are required to evaluate the evolution limits of any reactive system—are fundamentally performed using either a reaction based stoichiometric (S) approach or a non-stoichiometric (NS) method. One objective of the present paper is to check and discuss whether both methods lead to comparable solutions in the case of complex chemical equilibria encountered in high temperature and pressure combustion of hydrazine and propane and in steam gasification of glucose and cellulose. Within both calculation models various numerical methods can be used. We apply several techniques for each approach and discuss their easiness of implementation and usage, in particular when chemical equilibria are computed for reacting systems under a multitude of temperature and pressure conditions and feed compositions. In addition, we present a novel approach to perform sensitivity analyses based on a combination of S or NS methods and arc-length continuation technique. For this purpose, three industrially relevant case studies are exposed: (1) synthesis of ammonia using the Haber process, (2) steam gasification of a typical biomass surrogate, glucose, and (3) steam gasification of cellulose. For all the above reacting systems, our results are benchmarked against their counterparts obtained either from the ubiquitous process simulator: ASPEN-Plus® or from data available in the open literature.

KEYWORDS

arc-length continuation, cellulose, complex chemical reaction equilibria, glucose, Haber synthesis, hydrazine combustion, MATHEMATICA® and ASPEN-PLUS®, propane combustion, steam gasification, stoichiometric and non-stoichiometric methods, syngas production

JEL CLASSIFICATION

Chemical engineering

1 | INTRODUCTION

Calculation of complex chemical equilibria has been for decades—and remains in some cases even nowadays—a huge challenge. The fields where accurate knowledge of chemical equilibria is necessary encompass high temperature and pressure reactions, design of explosives, development of rocket propellants, design, and conception of new chemical processes, combustion, biomass gasification and so forth.

Calculations of chemical equilibria are performed in order to^{1–3}

- Determine the equilibrium composition of mixtures (mole or mass fractions, concentrations, partial pressures, etc.) for fixed operating conditions (e.g., reactants' composition, temperature and pressure), or/and
- Investigate the evolution of the equilibrium compositions when the operating conditions are varied (feed composition, temperature, pressure).

Fundamentally, two alternative thermodynamic methods can be used for the calculation of chemical equilibria at fixed feed composition, temperature, and pressure^{1–6}:

- Stoichiometric or S-approach. This procedure is best exemplified by the standard equilibrium constant method. With known nature and amounts of the reactants, a set of $\mathcal{R}$ chemical reactions are formulated relating reactants and chemical species present in the equilibrium mixture. These “reactions” do not necessarily represent actual chemical transformations but rather filiation routes. In fact, they just express the conservation of the chemical elements involved. The compositions of the equilibrium mixture are then determined by solving the set of equilibrium conditions for the different reactions using the corresponding equilibrium constants or, alternatively, by minimizing the Gibbs free energy with respect to the reaction extent variables.
- Non-stoichiometric or NS-approach. Here, no particular reactions are formulated. The amounts (number of moles) of the $\mathcal{N}$ species at equilibrium are determined by minimizing the Gibbs free energy G of the mixture subject to the constraints of the mass conservation equations of the chemical elements in the feed.

As the S-method is also based on minimization of the Gibbs energy of the mixture, it could be concluded that the results of both approaches are expected to be the same. However, it has been recently shown for the gasification of biomass that this is not always the case.⁴ For both methods to yield similar results, the number $\mathcal{R}$ of the considered reactions in the S-method should be so that

$$\mathcal{R} = \mathcal{N} - \text{rank}(\mathcal{A})$$

where $\mathcal{N}$ is the number of species present at equilibrium and $\mathcal{A}$ their formula matrix. Further, the selected $\mathcal{R}$ reactions must be linearly independent, or expressed in the context of linear algebra, the columns of the stoichiometric matrix $\bar{\nu}$ of the $\mathcal{R}$ reactions should form a basis for the null space of matrix $\mathcal{A}$.^{2,4}

$$\mathcal{A}\bar{\nu} = 0$$

One objective of the present paper is to check and discuss the equivalence or not-equivalence of S- and NS-approaches for a set of selected chemical systems: hydrazine and propane combustion on one side and glucose and cellulose steam gasification on the other side.

Different mathematical and numerical techniques are available for the calculations of chemical equilibria at fixed initial values of reactants' composition, temperature, and pressure.^{1–3} We can mention at first that all the techniques for the resolution of a set of non-linear algebraic equations are not only suitable in the case of the S-method in combination with equilibrium constants, but also for the NS-method where Lagrangian multipliers are applied for equality constraints. The constrained minimization problem is then reduced to a set of nonlinear algebraic equations.

A second group of numerical methods encompass all optimization techniques where unconstrained minimization of Gibbs free energy is used to determine the composition of the equilibrium mixture in the case of the NS-method or the reaction extents of the selected reactions in the case of the S-method.

Thus, a large variety of numerical procedures are available for the calculation of complex chemical equilibria within each of both approaches. A *second objective* of the present paper is to compare the easiness of implementation and use of some of these techniques and their accuracy in the prediction of the equilibrium. Hydrazine and propane combustion at high temperature and pressure are selected to exemplify both equilibrium calculation paradigms by using various numerical methods. For the sensitivity analyses of the equilibrium state when one of the following parameters—feed composition, temperature or pressure—is varied, we study the following three cases: the classical Haber ammonia process on one hand, and cellulose and glucose steam gasification, on the other hand. Arc-length continuation technique^{7,8} is applied to this purpose, a procedure that, to our best knowledge, has not yet been exploited for the calculation of chemical equilibria.

A *further objective* of the present paper is to show, whenever experimental data are available in the open literature, that classical thermodynamic equilibria with relatively simple equations of state (ideal gas law, Peng-Robinson equation of state) can indeed accurately predict real world data.

Three appendices provide the reader with some background information on useful chemical engineering thermodynamics and the arc-length procedure. In Appendix A, we give a concise description of the Peng-Robinson equation of state (PR-EoS) used in the sensitivity analyses to describe the volumetric behavior of gas mixtures under high pressure and to deduce the corresponding residual Gibbs free energy of such systems. Appendix B gives a full description of the procedure and the application of arc-length continuation in order to perform various parametric studies for chemically reacting systems. Appendix C finally, presents the Gibbs energy minimization procedure used in the NS-approach for the equilibrium calculation.

2 | COMPLEX CHEMICAL EQUILIBRIA CALCULATION METHODS

In the following two case studies of hydrazine and propane combustion, both approaches of equilibrium calculation, S- and NS-methods, are used and their results compared and discussed.

For each approach different numerical solution methods are applied.

2.1 | Hydrazine combustion

Hydrazine, with the chemical formula N_2H_4 , is primarily used as foaming agent in the preparation of polymer foams. It is also employed to reduce the concentration of dissolved oxygen, a source of corrosion, in the boilers of large conventional or nuclear power plants. A further application of hydrazine is as rocket fuel. Rocket engines are often fueled with this monopropellant in the last descent step of spacecraft (e.g., Curiosity and Perseverance rover landing on Mars in 2012 and February 2021, respectively). In these cases, hydrazine is catalytically decomposed into its elements: hydrogen and nitrogen and some ammonia. This exothermic reaction produces a huge amount of gas, and hence high pressure, from a small volume of liquid.

The present illustration does not investigate the catalytic decomposition of hydrazine but rather its combustion with pure oxygen at high temperature (3500 K) and pressure (51 bar). The gaseous product mixture involves many species from molecular compounds to atomic elements and radicals. In Reference 9, the product mixture equilibrium composition of this reaction involving 10 species (H, H_2 , N, N_2 , O, O_2 , H_2O , NO, NH, and OH) was calculated using the minimization of the Gibbs free energy. Two numerical procedures were applied for the solution: steepest descent and linear programming.

In the following, we include two more species, NO_2 and NH_2 , to the system and solve the problem applying three different methods using MATHEMATICA®.

2.1.1 | NS-Method: Lagrangian multipliers and element potentials

Assuming ideal gas behavior of the reacting mixture, the problem at hand consists of finding for fixed T and P , the mole numbers of the twelve species n_i that minimize the Gibbs free energy of the gaseous mixture $G_M(T, P, n_i)$,

$$G_M(T, P, n_i) = G_M(T, P^0, n_i) + RT \ln \left(\frac{P}{P^0} \right) \quad (1)$$

$G_M(T, P^0, n_i)$ is the Gibbs free energy at temperature T and reference pressure P^0 (e.g., 1 bar),

$$G_M(T, P^0, n_i) = \sum_{i=1}^{12} n_i \mu_i^0(T, P^0) \quad (2)$$

where $\mu_i^0(T, P^0)$ is the chemical potential of specie i at T and P^0 ,

$$\mu_i^0(T, P^0) = g_i^0(T, P^0) + RT \ln(y_i) \quad (3)$$

$g_i^0(T, P^0)$ is the molar free energy of the pure species i and y_i is its molar fraction in the mixture.

The solution mole numbers n_i minimizing G_M are subject to the constraints of the element-abundance equations. These constraints write:

- Oxygen balance

$$\mathcal{F}_O = n_O^0 - (y_O + 2y_{O_2} + y_{NO} + y_{H_2O} + y_{OH} + 2y_{NO_2})n = 0 \quad (4)$$

with

$$n = \sum_{i=1}^{12} n_i \quad (5)$$

- Hydrogen balance

$$\mathcal{F}_H = n_H^0 - (y_H + 2y_{H_2} + y_{NH} + 2y_{H_2O} + y_{OH} + 2y_{NH_2})n = 0 \quad (6)$$

- Nitrogen balance

$$\mathcal{F}_N = n_N^0 - (y_N + 2y_{N_2} + y_{NH} + y_{NO} + y_{NO_2} + y_{NH_2})n = 0 \quad (7)$$

where n_O^0 , n_H^0 , and n_N^0 are the initial amounts of the elements oxygen, hydrogen and nitrogen in the feed. In the case of a stoichiometric ratio of the reactants (hydrazine and oxygen), the initial mole numbers are then

$$n_O^0 = 2; n_H^0 = 4; n_N^0 = 2$$

The Lagrangian function of the mixture can be expressed as

$$\mathcal{L} = G_M(T, P, n_i) - \lambda_O \mathcal{F}_O - \lambda_H \mathcal{F}_H - \lambda_N \mathcal{F}_N \quad (8)$$

λ_O , λ_H , and λ_N are the Lagrange multipliers. We obtain by minimizing $\mathcal{L}$ in respect to n_i the following set of very interesting expressions for the chemical potentials of the mixture components at equilibrium (Table 1),

TABLE 1 Relations between species chemical potentials and element potentials

$\mu_O(T, P, n_i) = \lambda_O$	$\mu_{H_2O}(T, P, n_i) = 2\lambda_H + \lambda_O$
$\mu_{O_2}(T, P, n_i) = 2\lambda_O$	$\mu_{NH}(T, P, n_i) = \lambda_N + \lambda_H$
$\mu_H(T, P, n_i) = \lambda_H$	$\mu_{NH_2}(T, P, n_i) = \lambda_N + 2\lambda_H$
$\mu_{H_2}(T, P, n_i) = 2\lambda_H$	$\mu_{NO_2}(T, P, n_i) = \lambda_N + 2\lambda_O$
$\mu_N(T, P, n_i) = \lambda_N$	$\mu_{NO}(T, P, n_i) = \lambda_N + \lambda_O$
$\mu_{N_2}(T, P, n_i) = 2\lambda_N$	$\mu_{OH}(T, P, n_i) = \lambda_O + \lambda_H$

These relations suggest that the Lagrange multipliers can be considered as element potentials,¹⁰ for the chemical potential of every compound is equal to the sum of the potentials of its constituting elements affected by factors corresponding to its chemical formula. Further, these element potentials—for each chemical element there is only one potential—together with the total number of moles of the mixture are the only unknowns of the problem, regardless the number of chemical species present.

We can now express the mole fraction of each of these species as function of the appropriate Lagrange multipliers, for instance, for H₂O and OH:

$$y_{\text{H}_2\text{O}} = \exp \left[\frac{-g_{\text{H}_2\text{O}}(T, P) + 2\lambda_{\text{H}} + \lambda_{\text{O}}}{RT} \right] \quad (9)$$

or

$$y_{\text{OH}} = \exp \left[\frac{-g_{\text{OH}}(T, P) + \lambda_{\text{H}} + \lambda_{\text{O}}}{RT} \right] \quad (10)$$

After replacing the mole fractions by their expressions in the linear constraint relations and adding the summation equation:

$$\sum_{i=1}^{12} y_i = 1 \quad (11)$$

We obtain a set of *only* four algebraic equations in four variables (λ_{H} , λ_{O} , λ_{N} , and n) that can readily be solved using the MATHEMATICA[®] function FindRoot. The results for $T = 3500$ K and $P = 51$ bar are given in Table 2. In the last column, we give, for the sake of comparison, the compositions reported in Reference 9 obtained at the same conditions. As already mentioned, NH₂ and NO₂ were not included in the gaseous mixture considered in that reference; their actual compositions are effectively very small, 3.92×10^{-6} and 10^{-5} respectively, so that they have little effect on the mole fractions of the remaining species. For 8 of the 10 mixture components the results of the present calculations are in good agreement with those of Reference 9. For the species N and NH, the noticed discrepancy is attributed to the more recent values of $g_i^{\circ}(T, P^{\circ})$ used here, calculated basing on enthalpy and entropy formulae from NIST webbook.¹¹

2.1.2 | S-method

The same problem can be solved using the traditional equilibrium constant method. Possible routes to the product species involved in the hydrazine combustion process might be the reactions 1–9 in Table 3.

TABLE 2 Equilibrium composition of the product gas for hydrazine combustion

Species	y_i	$g_i^{\circ}(T, P^{\circ}) / RT$	y_i^9
N	0.00002	−6.06608	0.00086
O	0.01010	−14.7444	0.01095
H	0.02465	−10.1769	0.02482
NH	4.35×10^{-6}	−14.6569	0.00042
NO	0.01627	−28.3447	0.01672
NH ₂	3.92×10^{-6}	−24.501	-
H ₂ O	0.47846	−38.4983	0.47800
O ₂	0.02156	−30.9105	0.02277
N ₂	0.29731	−28.9657	0.29617
H ₂	0.08867	−21.4048	0.09015
OH	0.06295	−26.5222	0.05913
NO ₂	0.00001	−36.6686	-

The equilibrium constant $K_j(T)$ of a reaction (j) is related to its molar reaction Gibbs energy $\Delta_j G^0(T)$ at standard pressure P^0 as follows

$$K_j(T) = \exp \left[-\frac{\Delta_j G^0(T, P^0)}{RT} \right] \quad (12)$$

with

$$\Delta_j G^0(T, P^0) = \sum_i \nu_{ij} g_i^0(T, P^0) \quad (13)$$

where ν_{ij} denotes the stoichiometric coefficient of compound i in reaction j , with the usual sign convention: ν_{ij} is positive for products and negative for reactants.

Values of equilibrium constants for seven of the considered reactions (reactions 1–7 in Table 3) are tabulated as function of temperature in the literature, for example, Reference 12. For the remaining two reactions (8 and 9), the equilibrium constants can be evaluated using the corresponding NIST-Webbook¹¹ enthalpy and entropy temperature functions or the tabulated Gibbs free energy versus temperature from the literature for example, References 13–15.

The problem at hand is again to determine the molar fractions of the twelve species as well as the total number of moles n of the gas mixture. To solve for the 13 unknowns we join the three element mass balance equations and the summation equation to the nine equations of column 2 of Table 3. The set of the 13 algebraic equations is then solved applying the iterative Gauss–Newton algorithm implemented in the MATHEMATICA[®] command FindRoot.

As can be noticed from Table 4, we obtain almost the same results with both equilibrium calculation approaches: Gibbs free energy minimization and equilibrium constants method.

The solution for the stoichiometric method can alternatively be attained using a minimization procedure such as that of Levenberg–Marquardt for unconstrained minimization of the objective function $\mathcal{G}$ defined as follows:

$$\begin{aligned} \mathcal{G} = & \left(K_1(T) - \frac{y_{\text{H}}^2}{y_{\text{H}_2}} P \right)^2 + \left(K_2(T) - \frac{y_{\text{O}}^2}{y_{\text{O}_2}} P \right)^2 + \left(K_3(T) - \frac{y_{\text{N}}^2}{y_{\text{N}_2}} P \right)^2 + \left(K_4(T) - \frac{y_{\text{H}_2}^2 y_{\text{O}_2}}{y_{\text{H}_2\text{O}}^2} P \right)^2 + \left(K_5(T) - \frac{y_{\text{OH}}^2 y_{\text{H}_2}}{y_{\text{H}_2\text{O}}^2} P \right)^2 \\ & + \left(K_6(T) - \frac{y_{\text{NO}}^2}{y_{\text{N}_2} y_{\text{O}_2}} \right)^2 + \left(K_7(T) - \frac{y_{\text{NO}_2}^2}{y_{\text{N}_2} y_{\text{O}_2}^2} P \right)^2 + \left(K_8(T) - \frac{y_{\text{NH}}^2}{y_{\text{N}_2} y_{\text{H}_2}} \right)^2 + \left(K_9(T) - \frac{y_{\text{NH}_2}}{\sqrt{y_{\text{N}_2} y_{\text{H}_2}} \sqrt{P}} \right)^2 + \mathcal{F}_{\text{O}}^2 \\ & + \mathcal{F}_{\text{H}}^2 + \mathcal{F}_{\text{N}}^2 + \left(\sum_{i=1}^{12} y_i - 1 \right)^2 \end{aligned} \quad (14)$$

TABLE 3 Reactions and equilibrium constants (hydrazine combustion)

R_i	Reaction	Equilibrium constant $K_j(T, P)$
R_1	$\text{H}_2 \rightleftharpoons 2 \text{H}$	$K_1(T) = \frac{y_{\text{H}}^2}{y_{\text{H}_2}} P$
R_2	$\text{O}_2 \rightleftharpoons 2 \text{O}$	$K_2(T) = \frac{y_{\text{O}}^2}{y_{\text{O}_2}} P$
R_3	$\text{N}_2 \rightleftharpoons 2 \text{N}$	$K_3(T) = \frac{y_{\text{N}}^2}{y_{\text{N}_2}} P$
R_4	$2 \text{H}_2\text{O} \rightleftharpoons 2 \text{H}_2 + \text{O}_2$	$K_4(T) = \frac{y_{\text{H}_2}^2 y_{\text{O}_2}}{y_{\text{H}_2\text{O}}^2} P$
R_5	$2 \text{H}_2\text{O} \rightleftharpoons \text{H}_2 + 2 \text{OH}$	$K_5(T) = \frac{y_{\text{OH}}^2 y_{\text{H}_2}}{y_{\text{H}_2\text{O}}^2} P$
R_6	$\text{N}_2 + \text{O}_2 \rightleftharpoons 2 \text{NO}$	$K_6(T) = \frac{y_{\text{NO}}^2}{y_{\text{N}_2} y_{\text{O}_2}}$
R_7	$\text{N}_2 + 2 \text{O}_2 \rightleftharpoons 2 \text{NO}_2$	$K_7(T) = \frac{y_{\text{NO}_2}^2}{y_{\text{N}_2} y_{\text{O}_2}^2} P$
R_8	$\text{N}_2 + \text{H}_2 \rightleftharpoons 2 \text{NH}$	$K_8(T) = \frac{y_{\text{NH}}^2}{y_{\text{N}_2} y_{\text{H}_2}}$
R_9	$\frac{1}{2} \text{N}_2 + \text{H}_2 \rightleftharpoons \text{NH}_2$	$K_9(T) = \frac{y_{\text{NH}_2}}{\sqrt{y_{\text{N}_2} y_{\text{H}_2}} \sqrt{P}}$

TABLE 4 Comparison of equilibrium composition calculated by different approaches (hydrazine combustion)

Species	Lagrange method	Equilibrium constant method	Unconstrained minimization method
N	0.00002	0.00002	0.00002
O	0.01010	0.01011	0.01012
H	0.02465	0.02463	0.02463
NH	4.35×10^{-6}	4.34×10^{-6}	4.34×10^{-6}
NO	0.01627	0.01637	0.01631
NH ₂	3.92×10^{-6}	3.91×10^{-6}	3.91×10^{-6}
H ₂ O	0.47846	0.47854	0.47854
O ₂	0.02156	0.02147	0.02147
N ₂	0.29731	0.29731	0.29730
H ₂	0.08867	0.08858	0.08858
OH	0.06295	0.06300	0.06301
NO ₂	0.00001	0.00001	0.00001

With the MATHEMATICA[®] command `FindMinimum` the set of molar fractions of the compounds that minimizes $\mathcal{G}$ is readily found. Indeed, a minimum of 10^{-23} for $\mathcal{G}$ is attained. Again, the obtained values of the mole fractions are identical with those of the equilibrium constant method, as columns 3 and 4 of Table 4 show.

2.1.3 | NS-method: Constrained minimization of Gibbs free-energy

Alternatively, the same problem can be solved using a direct minimization of the Gibbs free energy function

$$G_M(T, P, n_i) = n \sum_{i=1}^{12} y_i [\mu_i^0(T, P^0) + RT \ln P + RT \ln y_i] \quad (15)$$

subject to the 3 atom balance equations and the summation equation constraints.

In this case, the MATHEMATICA[®] command `FindMinimum` is used in association with the interior point algorithm. As Table 5 shows, almost the same values of the mole fractions are obtained as for the preceding solution methods.

2.1.4 | NS versus S-approach

Let us consider the stoichiometric method from the linear algebra perspective. All $N = 12$ species in the equilibrium mixture are combinations of the elements hydrogen, oxygen, and nitrogen. The formula matrix $\mathcal{A}$ of this chemical system writes.

	H	O	N	OH	NH	NO	H ₂	O ₂	N ₂	H ₂ O	NH ₂	NO ₂
H	1	0	0	1	1	0	2	0	0	2	2	0
O	0	1	0	1	0	1	0	2	0	1	0	2
N	0	0	1	0	1	1	0	0	2	0	1	1

with $\text{rank}(\mathcal{A}) = 3$. The vector $\mathbf{n}$ of mole numbers of these species at equilibrium

$$\mathbf{n}^T = \{n_N, n_O, n_H, n_{OH}, n_{NH}, n_{NO}, n_{H_2}, n_{O_2}, n_{N_2}, n_{H_2O}, n_{NH_2}, n_{NO_2}\}$$

TABLE 5 Results of the constrained minimization method in comparison with the previous solution techniques (hydrazine combustion)

Species	Lagrange method	Unconstrained minimization method	Constrained minimization method
N	0.00002	0.00002	0.00002
O	0.01010	0.01012	0.01014
H	0.02465	0.02463	0.02475
NH	4.35×10^{-6}	4.34×10^{-6}	4.35×10^{-6}
NO	0.01627	0.01631	0.01628
NH ₂	3.92×10^{-6}	3.91×10^{-6}	3.91×10^{-6}
H ₂ O	0.47846	0.47854	0.47808
O ₂	0.02156	0.02147	0.02160
N ₂	0.29731	0.29730	0.29723
H ₂	0.08867	0.08858	0.08880
OH	0.06295	0.06301	0.06306
NO ₂	0.00001	0.00001	0.00001

has to satisfy the three elemental abundance equations

$$\mathcal{A}\mathbf{n}^T = \begin{Bmatrix} n_{\text{N}}^0 \\ n_{\text{O}}^0 \\ n_{\text{H}}^0 \end{Bmatrix}$$

and the nine equilibrium conditions for the reactions of Table 3. According to Reference 4, the NS and S equilibrium calculation approaches would yield concordant results if for this system, the number $\mathcal{R}$ of the considered set of *independent* reactions in the S-method is related to the number $\mathcal{N}$ of species and the rank of the formula matrix by the expression

$$\mathcal{R} = \mathcal{N} - \text{rank}(\mathcal{A})$$

The stoichiometric (9×12)-matrix of the 9 reactions of Table 3 writes.

	H	O	N	OH	NH	NO	H2	O2	N2	H2O	NH2	NO2
R1	2	0	0	0	0	0	-1	0	0	0	0	0
R2	0	2	0	0	0	0	0	-1	0	0	0	0
R3	0	0	2	0	0	0	0	0	-1	0	0	0
R4	0	0	0	0	0	0	2	1	0	-2	0	0
R5	0	0	0	2	0	0	1	0	0	-2	0	0
R6	0	0	0	0	0	2	0	-1	-1	0	0	0
R7	0	0	0	0	0	0	0	-2	-1	0	0	2
R8	0	0	0	0	2	0	-1	0	-1	0	0	0
R9	0	0	0	0	0	0	-1	0	$-\frac{1}{2}$	0	1	0

It can be easily tested that the rank of this matrix is 9, that is, the rows of this matrix are independent and so also are the reactions $R_1 - R_9$. In summary we have ($\mathcal{N} = 12, \text{rank}(\mathcal{A}) = 3, \mathcal{R} = 9$) and thus, the equivalence condition of the stoichiometric and non-stoichiometric approaches is established. And indeed, as already noticed, Tables 4 and 5 show that the results of both methods are indeed concordant.

2.2 | Propane combustion: Equilibrium calculation using NS-methodology

Combustion of hydrocarbon fuels with air produces a gaseous mixture composed at least of the following thirteen species: O_2 , O , H_2 , H_2O , OH , N_2 , NO , CO_2 , CO , C , H , N , NO_2 .¹⁶ We consider the particular case of the combustion of propane with a stoichiometric amount of air, assumedly composed of 21 vol.% oxygen and 79 vol.% nitrogen. The equilibrium composition of the gaseous product can be easily found using the element potential method (NS-approach).

The Lagrangian function of the mixture writes

$$\mathcal{L} = G_M(T, P, n_i) - \lambda_O \mathcal{F}_O - \lambda_H \mathcal{F}_H - \lambda_N \mathcal{F}_N - \lambda_C \mathcal{F}_C \quad (16)$$

with

$$G_M(T, P, n_i) = n \sum_{i=1}^{13} y_i [\mu_i^0(T, P^0) + RT \ln P + RT \ln y_i] \quad (17)$$

where $\mathcal{F}_O$, $\mathcal{F}_H$, $\mathcal{F}_N$, and $\mathcal{F}_C$ express the atom balance constraints, respectively.

$$\mathcal{F}_O = 10 - (y_O + 2y_{O_2} + y_{NO} + y_{H_2O} + y_{OH} + 2y_{NO_2} + 2y_{CO_2} + y_{CO})n = 0 \quad (18)$$

$$\mathcal{F}_H = 8 - (y_H + 2y_{H_2O} + 2y_{H_2} + y_{OH})n = 0 \quad (19)$$

$$\mathcal{F}_N = 10 \left(\frac{79}{21} \right) - (y_N + 2y_{N_2} + y_{NO} + y_{NO_2})n = 0 \quad (20)$$

$$\mathcal{F}_C = 3 - (y_C + y_{CO} + y_{CO_2})n = 0 \quad (21)$$

with

$$n = \sum_{i=1}^{13} n_i \quad (22)$$

Similarly to the hydrazine combustion case, we solve now a set of just five algebraic equations (the atom balance equations and the summation equation) for the Lagrange multipliers λ_O , λ_H , λ_N , λ_C and the total molar amount of the mixture, n . Columns 2 and 3 of Table 6 show the composition of the gas mixture obtained using the present method in comparison with the NASA NS-method¹⁷ where an iterative successive substitution algorithm involving matrix inversion is applied to minimize the Gibbs free-energy.

As can be noticed the agreement between both methods is almost perfect.

In Reference 16 the same problem was solved using a stoichiometric approach. To this purpose, nine independent reactions are considered (Table 7).

The formula matrix of the equilibrium mixture is,

	N	O	H	C	OH	CO	NO	H2	O2	N2	H2O	CO2	NO2
N	1	0	0	0	0	0	1	0	0	2	0	0	1
O	0	1	0	0	1	1	1	0	2	0	1	2	2
H	0	0	1	0	1	0	0	2	0	0	2	0	0
C	0	0	0	1	0	1	0	0	0	0	0	1	0

Its rank is 4. Hence, nine independent reactions are needed to ensure the equivalence of the S and NS-approach, for instance the reactions collection of Table 7. As noticed, the mixture composition calculated using this S-approach and given in the last column of Table 6 is almost identical with those in the second and third columns of the same table evaluated adopting the NS-method. Incidentally, the nullspace of the formulae matrix generate exactly the same reactions set as the one given in Table 7.

TABLE 6 Equilibrium composition of the product gas for propane combustion

Species	Lagrange method	NASA ¹⁷	S-Approach ¹⁶
O ₂	0.00402	0.00404	0.00402
O	0.00007	0.00007	0.00007
H ₂	0.00015	0.00015	0.00015
H ₂ O	0.09638	0.09640	0.09638
OH	0.00148	0.00143	0.00148
N ₂	0.71931	0.71931	0.71932
NO	0.00320	0.00320	0.00319
CO ₂	0.16621	0.16624	0.16622
CO	0.00917	0.00915	0.00916
C	2.2265×10^{-16}	<0.00001	<0.00001
H	5.8416×10^{-6}	<0.00001	<0.00001
N	1.7339×10^{-8}	<0.00001	<0.00001
NO ₂	3.6451×10^{-6}	<0.00001	<0.00001

TABLE 7 Reactions set adopted in Reference 16

R_i	Reaction	R_i	Reaction
R_1	$H_2 \rightleftharpoons 2 H$	R_6	$CO \rightleftharpoons C + O$
R_2	$O_2 \rightleftharpoons 2 O$	R_7	$CO_2 \rightleftharpoons C + 2 O$
R_3	$N_2 \rightleftharpoons 2 N$	R_8	$NO \rightleftharpoons N + O$
R_4	$H_2O \rightleftharpoons 2 H + O$	R_9	$NO_2 \rightleftharpoons N + 2 O$
R_5	$OH \rightleftharpoons H + O$		

2.3 | Concluding remarks of Section 2

From the study of hydrazine and propane combustion we can conclude that

- When the condition for the equivalence of the stoichiometric and non-stoichiometric approaches is fulfilled, both NS and S calculation models lead to same results.
- In the considered case studies, and from practical point of view, the NS-method in combination with the Lagrangian multipliers is easier to implement and to use than other methods because of the much fewer number of unknowns to solve for and consequently the number of initial values to start the iteration procedure.

3 | SENSITIVITY ANALYSES

3.1 | Haber synthesis

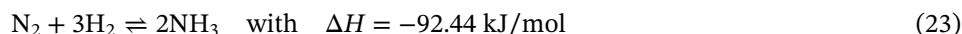
Ammonia is an essential compound to all biological processes because nitrogen is present in all proteins and related biological molecules. It is also of strategic importance as it is used as refrigerant in cold producing machines and to produce fertilizers and explosives. The world's production of ammonia in 2019 capped at 235 million tons and it is expected to reach 290 million tons by 2030.¹⁸ We owe the catalytic synthesis of ammonia to the Nobel Prize winner Fritz Haber. The fabrication process of ammonia is performed in two major steps: synthesis gas production and purification, and ammonia synthesis. The reactants, nitrogen and hydrogen, are produced from air and hydrogen containing hydrocarbons, for example, natural gas. Secondary air reforming of natural gas follows a primary steam reforming. Most of the obtained

carbon monoxide is transformed in carbon dioxide, which is then removed. The rest of CO undergoes a methanation process. The ultimate synthesis feedstock is composed of hydrogen and nitrogen as well some inert gases, typically methane and argon.

The industrial process is energy-intensive; it requires high pressures and produces greenhouse gases (i.e., CO₂). Hence, recent research work is directed toward the production of “green” ammonia under mild conditions. New routes to ammonia synthesis include thermo-catalytic, electro-catalytic, photo-catalytic, and chemical looping processes. Wang and co-workers provide a recent review of the state-of-the-art research in this area.¹⁹

We choose to focus in the present paper on the last step of the Haber-Bosch process for which the feedstock is a mixture of hydrogen and nitrogen.

The synthesis of ammonia is an exothermic reaction:



The reaction is also associated with a reduction in volume. The implications of these two characteristics of the reaction will be discussed in the text thereafter. The values of the PR EoS binary interaction parameters for this system are provided in Table 8.

Denoting with r the initial molar ratio of the reactants, $r = \frac{n_{\text{H}_2}^0}{n_{\text{N}_2}^0}$, where superscript 0 refers to the initial number of moles, the equality constraints expressing the atomic mass balances write

$$\text{N} : 2n_{\text{N}_2} + n_{\text{NH}_3} = 2 \quad (24)$$

$$\text{H} : 2n_{\text{H}_2} + 3n_{\text{NH}_3} = 2r \quad (25)$$

3.1.1 | Stoichiometric mixture of hydrogen and nitrogen

We consider first the stoichiometric case, that is, $r = 3$. Figure 1 shows the effect of temperature and pressure on the molar fraction of ammonia in the product mixture. The continuous lines represent calculation results and the symbols, industrial data.²⁰ As can be noticed from Figure 1, the agreement between the two sets of data is excellent for all pressures and temperatures.

Figure 2 depicts the nitrogen conversion ratio

$$\chi = \frac{n_{\text{N}_2}^0 - n_{\text{N}_2}}{n_{\text{N}_2}^0} \quad (26)$$

for the same conditions. Here again, we notice the very good concordance of calculated and experimental data.²¹

In accordance with Le Chatelier's principle,²² the nitrogen conversion is (1) high at lower temperatures since the reaction is exothermic and (2) high for larger pressures since the reaction is associated with a volume reduction. Accordingly, in the industrial ammonia production processes, the pressure ranges frequently between 100 and 300 atm. Some processes are run even at pressures as high as 900 atm. The reaction temperature however is maintained rather high, ranging from 400°C to 450°C, to ensure high reaction rates.

The calculated results are obtained using a real gas equation of state, the Peng-Robinson EoS. Figure 3 depicts the compressibility factor of the reacting gas mixture at equilibrium for varying temperatures and pressures. It can be observed that the value of this factor in the considered range of temperature and pressure (1) does not deviate largely from that

TABLE 8 Peng–Robinson EoS interaction parameters for the Haber synthesis (obtained from Aspen-HYSYS)

	Hydrogen	Nitrogen	Ammonia
Hydrogen	0	−0.036	0
Nitrogen	−0.036	0	0.222
Ammonia	0	0.222	0

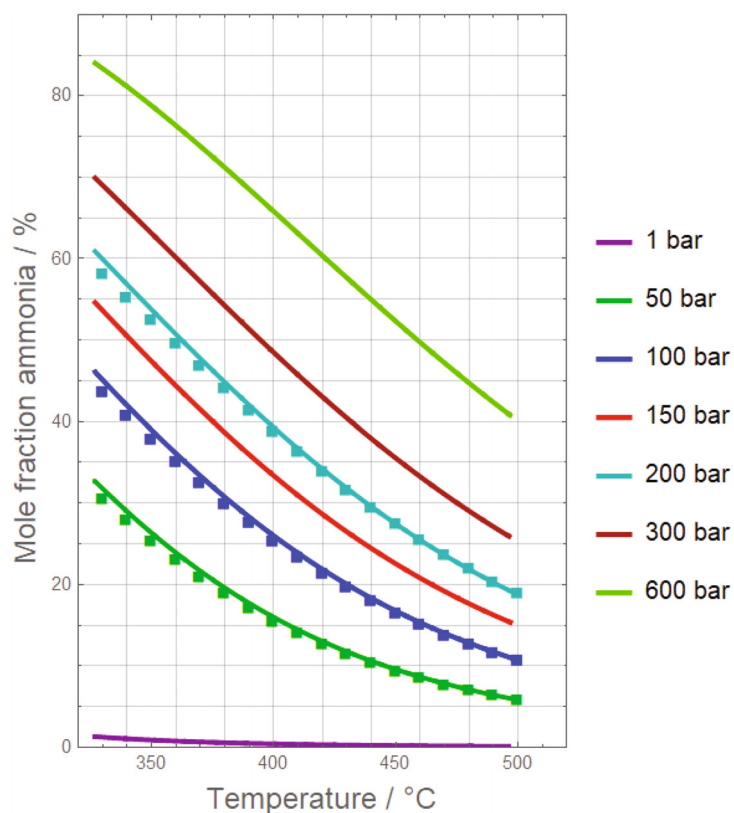


FIGURE 1 Equilibrium ammonia mole fraction versus temperature at various pressures for a gas containing initially a stoichiometric mixture of N_2 and H_2 . Continuous curves: Calculated using PR-EoS, Symbols: data²⁰

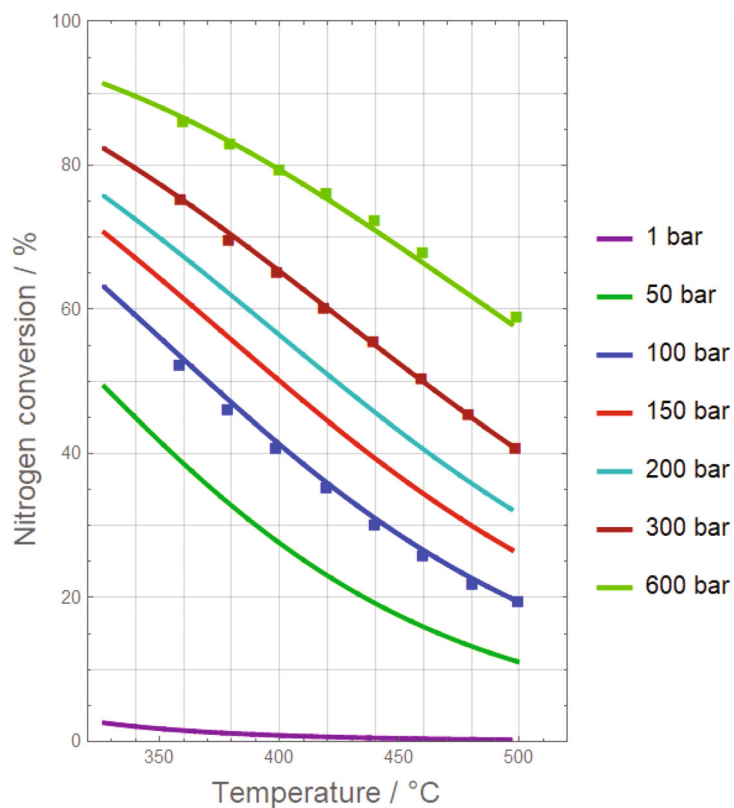


FIGURE 2 Equilibrium nitrogen conversion versus temperature at various pressures for a gas containing initially a stoichiometric mixture of N_2 and H_2 . Continuous curves: Calculated using PR-EoS, Symbols: data²¹

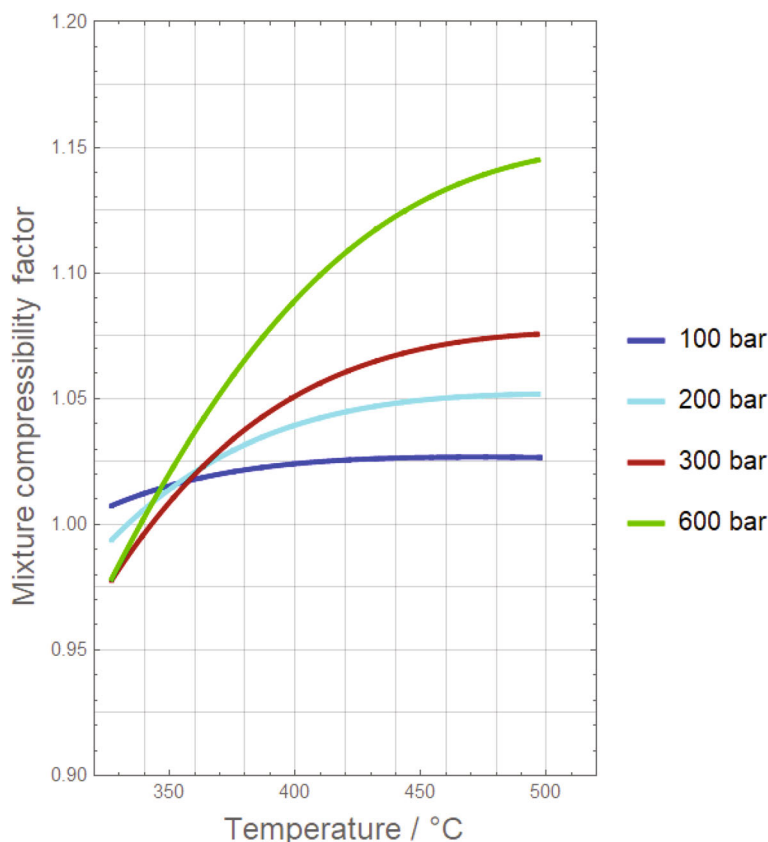


FIGURE 3 Mixture compressibility factor versus temperature at various pressures

of an ideal gas, that is, 1, and (2) increases with increasing temperature and pressure. For a pressure of 100 bars, the compressibility factor is almost constant and equals roughly 1.02 in the temperature range 350°C to 500°C. Under such conditions, the predicted nitrogen conversion ratio assuming ideal gas behavior is in fact in very good agreement with the experimental data as Figure 4 illustrates. For higher pressures (e.g., 300 and 600 bars), the predicted results using ideal gas assumption are very poor. Indeed, the deviation between such results and their counterparts using PR EoS become larger with increasing pressure (see Figure 4).

3.1.2 | Effect of process gas composition

Figure 5 shows the effect of temperature on the ammonia mole fraction when inert gases are present in the reactants stream. The calculations are performed for a pressure of 100 bars and assuming stoichiometric ratio of the reactants mixed with 10 mol.% inert gases: 7.5% methane and 2.5% argon. The results are benchmarked against industrial data²⁰ and ASPEN-Plus® simulation results. As can be noticed, all three sets of data are in good concordance.

As expected, the ammonia content of the equilibrium mixture is decreasing with higher temperature but is also decreased by the presence of inert gases, as the comparison with the results depicted in Figure 1 for the same conditions shows.

Figure 6 depicts the effect of process gas composition r for selected values of temperature by a fixed pressure $P = 150$ bars. We notice that for all temperatures the conversion first increases with increasing r , passes by a maximum for $r = 3$ and then decreases. Clearly, for $r < 3$, there is a deficit in hydrogen. The surplus of nitrogen acts as inert gas: the more of it, the lower the ammonia content in the equilibrium mixture. For $r > 3$, the conversion drops due to a deficit in nitrogen. It is now the excess of hydrogen that plays the role of an inert gas, and hence decreases the equilibrium content of the product ammonia.

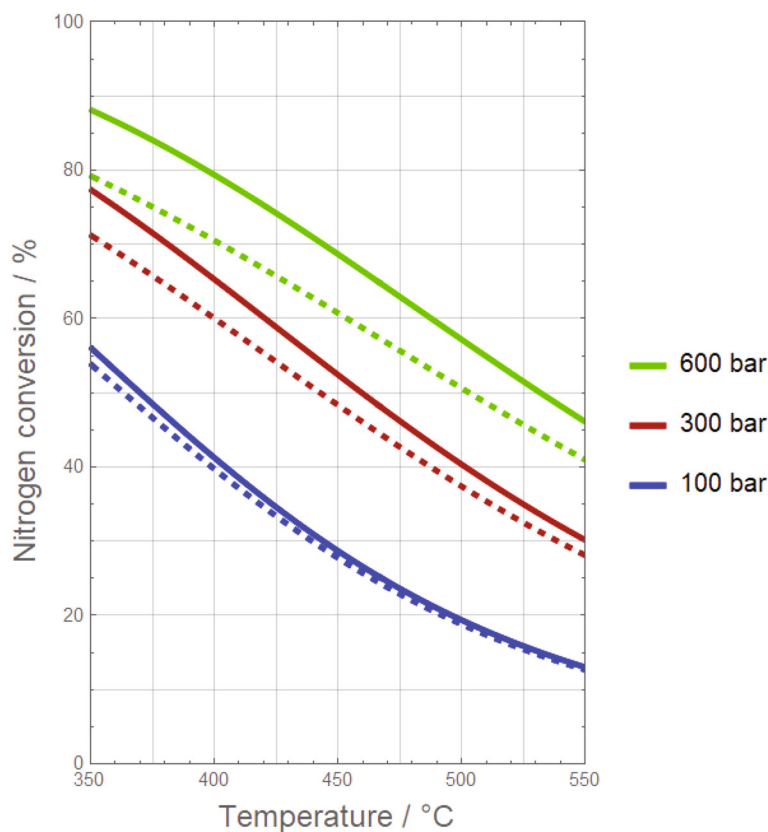


FIGURE 4 Calculated equilibrium conversion versus temperature at three different pressures assuming real gas behavior (continuous lines) and ideal gas behavior (dashed lines)

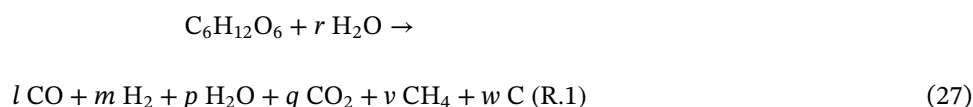
3.2 | Steam gasification of glucose

The transformation of biomass into energy-rich or useful products can be achieved via biological or thermochemical processes. However, literature screening shows that recently more attention has been focused on the thermochemical path, in particular on gasification.^{23,24} Biomass gasification is an interesting route to a sustainable energy and chemicals production since biomass is carbon-neutral, renewable, and abundant if compared to fossil fuels (gas, petroleum, and coal), which are being rapidly depleted.²⁵ The production of biogas, also called syngas, by gasification consists of partial oxidation of carbon containing biomass using gasifying agents such as air, steam, carbon dioxide and so forth. Generally, the produced syngas contains primordially carbon monoxide and hydrogen with minor amounts of methane, nitrogen, carbon dioxide, and lighter hydrocarbons. The produced syngas can then be used to generate electricity, liquid fuels, or chemical products.

As reported in Reference 4 a major part of the published investigations on biomass gasification models favor the equilibrium approach, the calculation of it being performed applying either the S or/and NS-methods.^{23–39}

The experimental study of glucose gasification using steam is of importance since glucose represents a simple biomass surrogate.³⁷ Here, we propose to perform thermodynamic computations of the equilibrium composition for this gasification process under various conditions. It should be noted however that the gasification involves heterogeneous reactions which are far from the thermodynamic equilibrium due to both low reaction rates and short residence times. Thus, thermodynamic predictions represent target values toward which the experimentalist should always aim.

The overall reaction between glucose and steam can be expressed as follows:



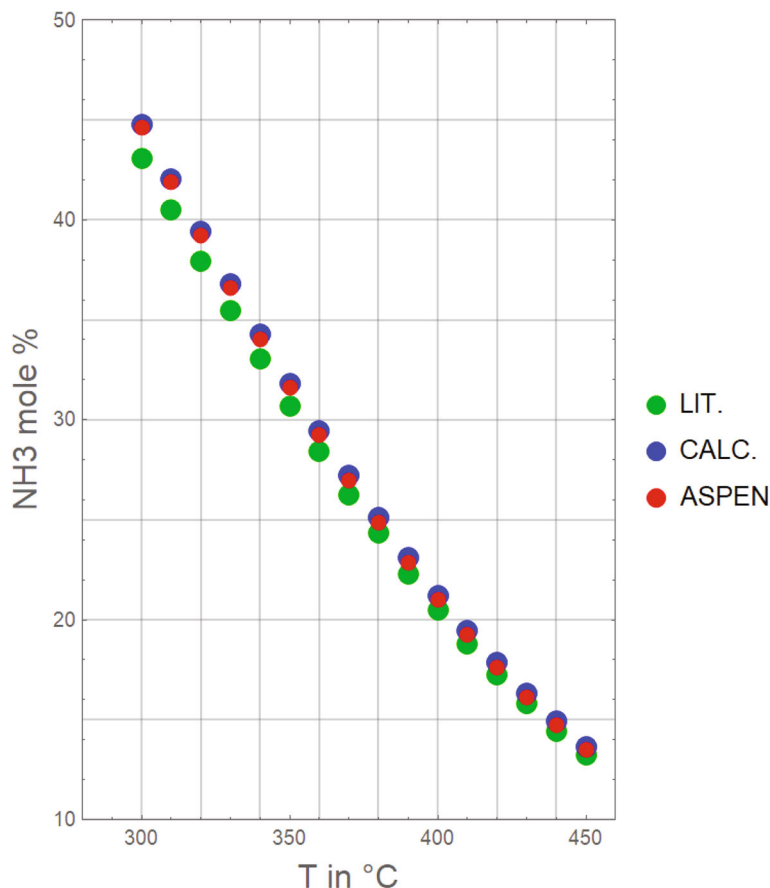


FIGURE 5 Equilibrium gas-phase composition versus temperature for a process gas for a stoichiometric feed mixture at 100 bars in the presence of inert gases (7.5 mol.% CH₄ and 2.5 mol.% Argon). Dots: Red, data obtained from RGIBBS of ASPEN-Plus®, Blue, calculated using PR-EoS, Green, literature data²⁰⁾

where r represents the steam/glucose molar ratio and

$$q = \left(3 + \frac{r-l-p}{2} \right); v = \left(3 + \frac{r-m-p}{2} \right); w = \left(p - r + \frac{m-l}{2} \right) \quad (28)$$

Table 9 lists some of the possible transformation routes to the gasification products.^{37,38} Among these reactions, we find the coke gasification (R.2), the Boudouard reaction (R.3), the hydrogenating reaction (R.4), the dry reforming of methane (R.5), the steam reforming of methane (R.6) and the water gas shift reaction (R.7).

For the equilibrium calculations of this system, we consider first the Gibbs free energy minimization approach (NS-method). The solid phase is supposed to be formed of graphite. Values of the PR EoS binary interaction parameters for commonly encountered gaseous compounds in glucose gasification using steam are given in Table 10 (obtained from Aspen-HYSYS).

Denoting with n_{CH_4} , n_{H_2} , $n_{\text{H}_2\text{O}}$, n_{CO} , n_{CO_2} , and n_{C} the number of moles at equilibrium of CH₄, H₂, H₂O, CO, CO₂, and C, respectively, the equality constraints write:

$$\text{C} : n_{\text{CH}_4} + n_{\text{CO}} + n_{\text{CO}_2} + n_{\text{C}} = 6 \quad (29)$$

$$\text{H} : 4n_{\text{CH}_4} + 2n_{\text{H}_2} + 2n_{\text{H}_2\text{O}} = 12 + 2r \quad (30)$$

$$\text{O} : n_{\text{H}_2\text{O}} + n_{\text{CO}} + 2n_{\text{CO}_2} = 6 + r \quad (31)$$

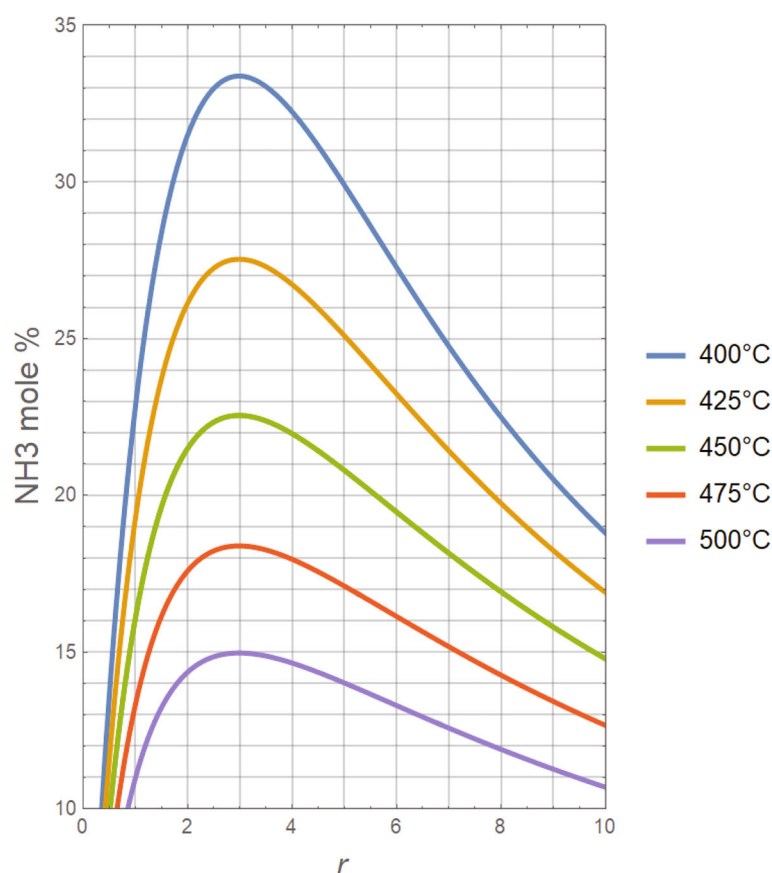


FIGURE 6 Equilibrium gas-phase composition for a process gas containing initially r moles of H_2 for 1 mole of N_2 at various temperatures for $P = 150$ bar

TABLE 9 Main reactions involved in the gasification process of glucose using steam

Chemical reaction		$\Delta_R H^0(298\text{ K})$ (kJ/mol)
$C + H_2O \rightleftharpoons H_2 + CO$	(R.2)	+131.4
$C + CO_2 \rightleftharpoons 2CO$	(R.3)	+172.6
$C + 2H_2 \rightleftharpoons CH_4$	(R.4)	-75.0
$CH_4 + CO_2 \rightleftharpoons 2CO + 2H_2$	(R.5)	+247
$CH_4 + H_2O \rightleftharpoons CO + 3H_2$	(R.6)	+201.9
$CO + H_2O \rightleftharpoons H_2 + CO_2$	(R.7)	-41.2

TABLE 10 Peng-Robinson EoS interaction parameters for biomass gasification products (obtained from Aspen-HYSYS)

	CH_4	H_2	H_2O	CO	CO_2
CH_4	0	0.2020	0.5000	0.0210	0.1000
H_2	0.2020	0	-0.2998	0.0253	0.1202
H_2O	0.5000	-0.2998	0	-0.3896	0.0445
CO	0.0210	0.0253	-0.3896	0	-0.0314
CO_2	0.1000	0.1202	0.0445	-0.0314	0

3.2.1 | Effect of pressure and temperature

Figures 7 to 9 illustrate the effect of pressure on the gasification equilibrium of glucose at $T = 1000$ K for an initial glucose to steam molar ratio equal to unity (i.e., $r = 1$). The composition of the gas phase versus pressure is depicted in Figure 7, the amount of solid carbon formed and the total number of moles in the gas phase are graphically represented in Figure 8 and the H_2/CO molar ratio, in Figure 9. The solid lines in Figure 7 represent the compositions of CH_4 , H_2 , H_2O , CO , and CO_2 obtained by our calculation using MATHEMATICA® while the colored dots are their counterparts gathered from RGIBBS of ASPEN-Plus®. Figures 7 and 8 show in accordance with Le Chatelier's principle²² that increasing the pressure disadvantages the expansive reactions (R.2, R.3, R.5, and R.6 in Table 9) and hence decreases the total number of moles in the gaseous mixture, promoting the formation of solid carbon and of the low-energy compounds H_2O and CO_2 . At the same time, the production of syngas (H_2 and CO) is largely hindered, even if the ratio H_2/CO is increased (Figure 9).

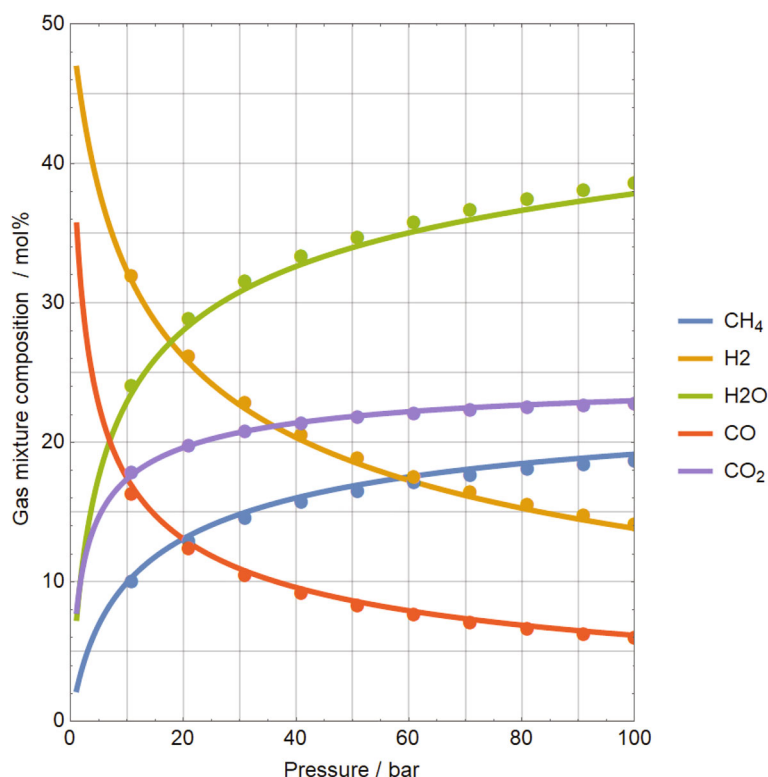


FIGURE 7 Equilibrium composition of the gas phase versus pressure an initial reaction mixture containing 1 mole glucose for 1 mole steam at $T = 1000$ K (solid lines: present calculation, dots: data computed by RGIBBS of ASPEN-Plus®)

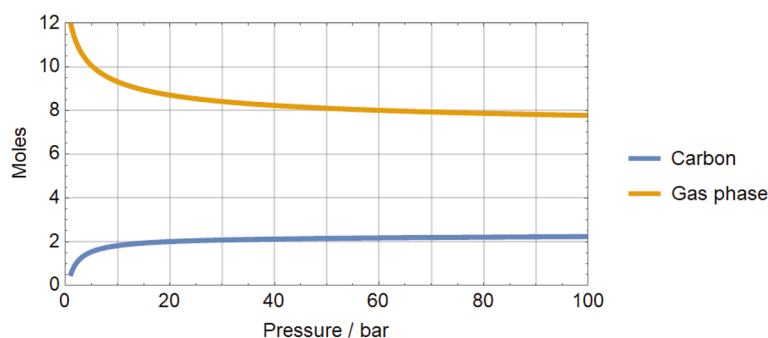


FIGURE 8 Total number of moles in gas phase and solid carbon amount versus pressure for an initial reaction mixture containing 1 mole glucose and 1 mole steam at $T = 1000$ K

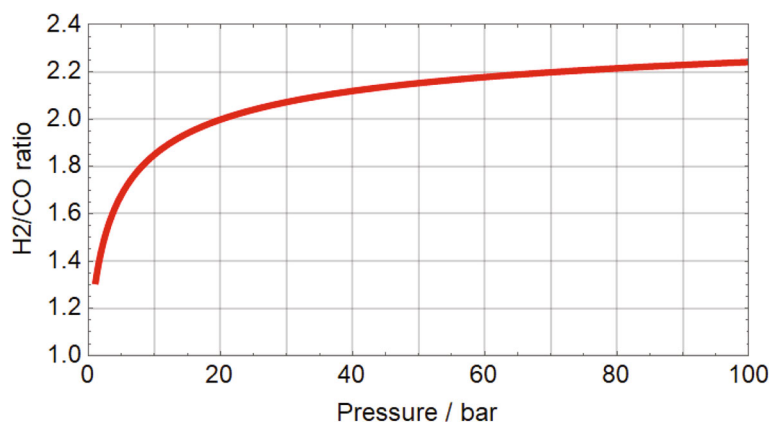


FIGURE 9 H_2/CO ratio versus pressure for an initial reaction mixture containing 1 mole glucose for 1 mole steam at $T = 1000$ K

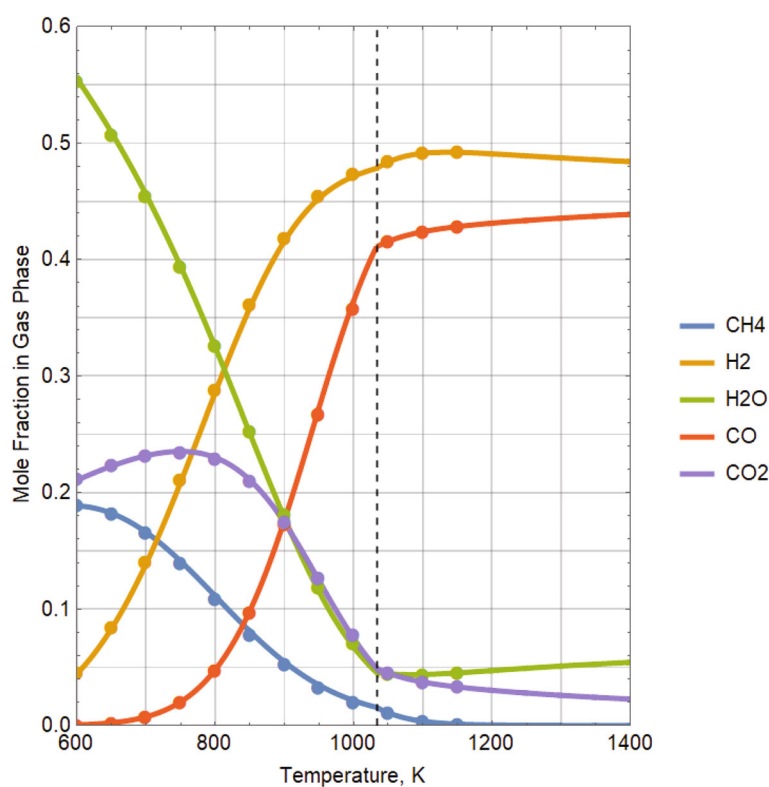


FIGURE 10 Equilibrium composition of the gas phase versus temperature for an initial reaction mixture containing 1 mole glucose for 1 mole steam at $P = 1$ bar (solid lines: present calculation, dots: data computed by RGIBBS of ASPEN-Plus®)

Figures 10 and 11 show the effect of temperature on the steam gasification equilibrium of glucose with a reactant molar ratio equal to unity and $P = 1$ bar. Again, the solid lines in Figure 10 represent the mole fractions of the different components in the gaseous phase obtained by our calculation while the colored squares the data obtained from RGIBBS of ASPEN-Plus®. Clearly, the agreement between both methods of calculation is excellent. Figure 11 depicts the evolution of the amount of solid carbon and the total number of moles in the gas phase. Low pressure and higher temperatures are now favoring the expansive and endothermic reactions (R.2, R.3, R.5, and R.6 in Table 9). Consequently,

- The total number of moles in the gas phase is increasing (Figure 11), and (2) the formation of the syngas components H_2 and CO is becoming predominant with increasing temperature (Figure 10).

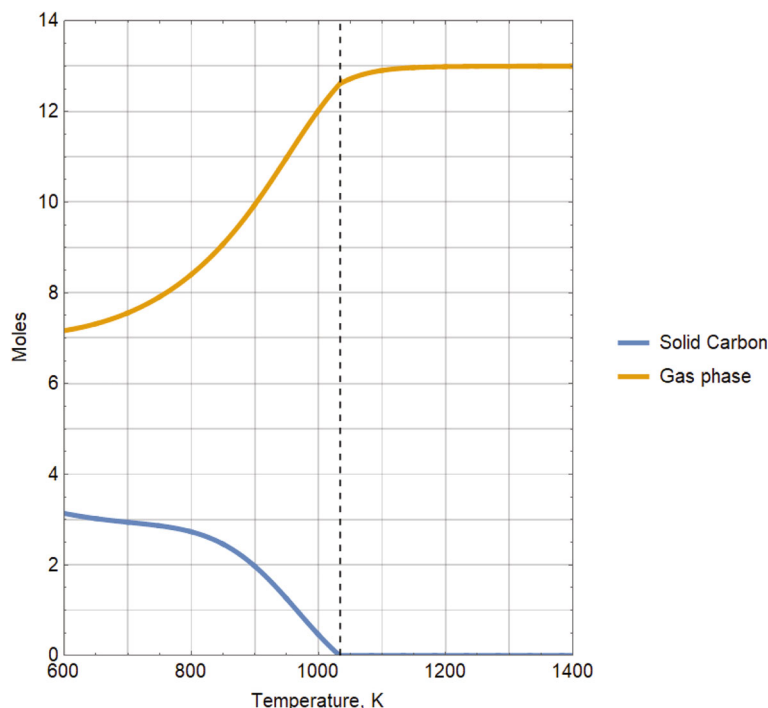


FIGURE 11 Solid carbon content and total number of moles in the gas phase versus temperature for an initial reaction mixture containing 1 mole glucose for 1 mole steam at $P = 1$ bar

- Further, the amount of solid carbon is rapidly decreasing (Figure 11) owing to the tar reforming reaction (R.2) and the Boudouard reaction (R.3). In fact, solid carbon is being continuously consumed with increasing temperature until it is totally depleted at $T \approx 760^\circ\text{C}$ (1033 K). The dashed vertical line in Figures 10 to 14 indicates this limit. From this temperature upwards, only homogeneous reactions R.5, R.6, and R.7 are involved.
- Concomitantly, the steam content of the gas mixture is also decreasing due to the strong endothermic reactions R.2 and R.6 where it is used as reactant.

The similar behavior of the curves representing the mole fractions of CO_2 and CH_4 in Figure 10 is attributable to the fact that these components are participating in the reacting mixture at the same time as reactants (R.3 and R.5 for CO_2 ; R.5 and R.6 for CH_4) and as products (R.7 for CO_2 ; R.4 for CH_4).

We now model the same equilibrium problem of the glucose steam gasification applying the reaction coordinate approach. The present chemical system involves 6 species (5 gaseous compounds and solid carbon) formed out of three chemical elements carbon, hydrogen and oxygen. The rank of the formula matrix of this system is three and the rank of the null-space matrix is also three. For the calculations we select therefore three linearly independent reactions from the reactions set given in Table 9, for example, reactions R.2, R.6, and R.7



Denoting by ξ_2 , ξ_6 , and ξ_7 their respective reaction extent variables, the amount n_i of each chemical species of the mixture is now expressed as

$$n_i = n_i^0 + \nu_{i,2}\xi_2 + \nu_{i,6}\xi_6 + \nu_{i,7}\xi_7$$

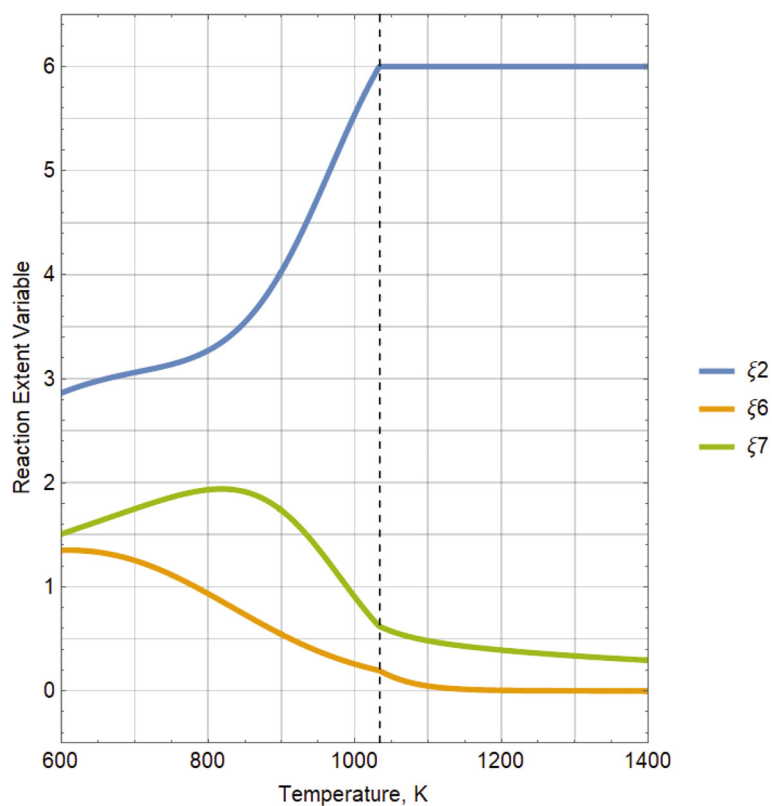


FIGURE 12 Reaction extent variables versus temperature for an initial reaction mixture containing 1 mole glucose for 1 mole steam at $P = 1$ bar

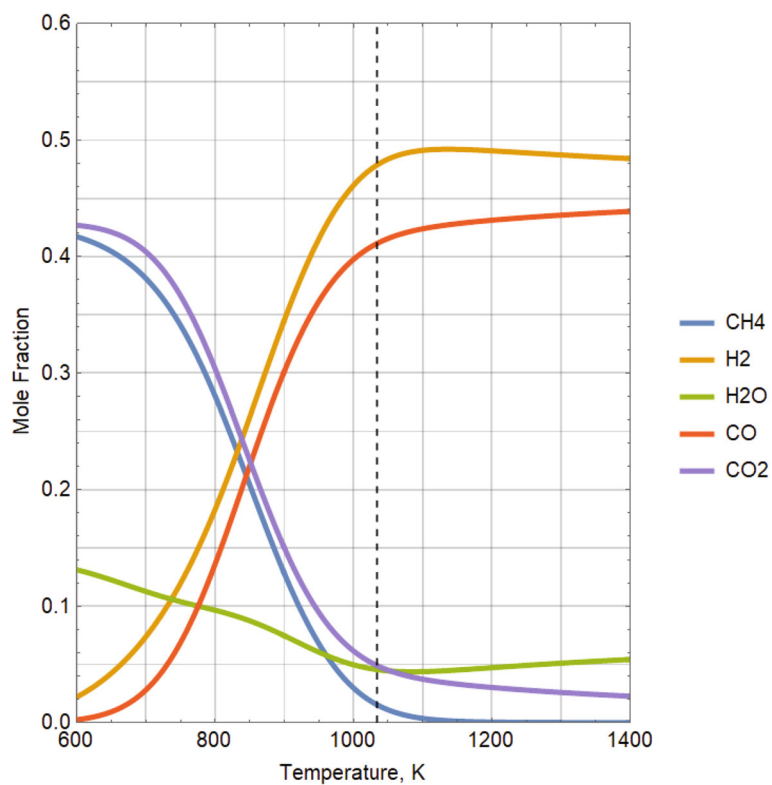


FIGURE 13 Equilibrium composition of the homogeneous gas phase versus temperature for an initial reaction mixture containing 1 mole glucose for 1 mole steam at $P = 1$ bar

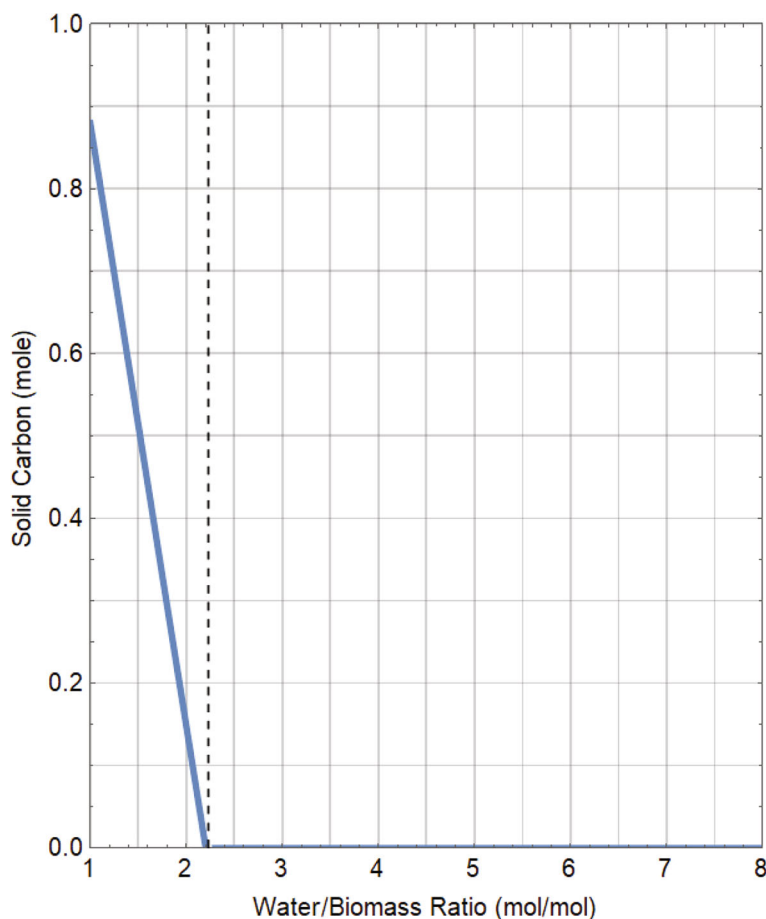


FIGURE 14 Effect of water/glucose-ratio on solid carbon content at $P = 1$ bar and $T = 973.15$ K (700°C)

The Gibbs free energy of the mixture is minimized with respect to the three variables ξ_2 , ξ_6 , and ξ_7 . Apart from the non-negativity constraints on the compound mole numbers, this is an unconstrained minimization problem that we solve using the MATHEMATICA[®] command `FindMinimum`. The composition of the equilibrium mixture obtained is identical to that found previously adopting the NS-approach and depicted in Figure 10. Once again, we notice the equivalence of S and NS models of equilibrium as long as the equivalence criterion for both approaches as expressed by the equation $\mathcal{R} = \mathcal{N} - \text{rank}(\mathcal{A})$ is fulfilled.

Figure 12 shows the evolution of the three reaction extent variables with temperature. In agreement with the preceding discussion of the NS equilibrium results, solid carbon is rapidly consumed with increasing temperature. From the limit temperature $T \approx 760^{\circ}\text{C}$ (1034 K) on, it is totally transformed.

If only gas-phase equilibria are considered—for example, solid carbon is supposed to have been totally vaporized and transformed during the pyrolysis step or separated—the equilibrium is determined by only two independent gaseous reactions,⁴ for instance reactions R.6 and R.7,



The equilibrium calculations, applying both S and NS models, yield concordant results depicted in Figure 13. Notice that these gas compositions are quite different from those of Figure 10 in presence of carbon, that is, for temperatures less than the limit of 1034 K. For higher temperatures, when all carbon is consumed, the compositions of the homogeneous gas phases are identical.

3.2.2 | Effect of glucose/steam ratio

Figures 14 and 15 depict the effect of the steam to glucose molar ratio, r , on the equilibrium amount of solid carbon and the compositions of the gas phase and at $T = 973.15$ K (700°C) and $P = 1$. Solid carbon is progressively depleted with increasing steam/glucose ratio due to the reforming reaction R.2. By a ratio of 2.23, all solid carbon is consumed. For larger values of r —that is, steady addition of steam and absence of carbon, a source for CH_4 , CO , and CO_2 —the gas phase chemical transformations R.5, R.6, and R.7 leads eventually to the exhaustion of methane and the continuous consumption of carbon monoxide in favor of carbon dioxide (water-shift reaction R.7). Concurrently, more hydrogen is produced. Figure 15 illustrates graphically these observations.

Figure 16 finally depicts the molar fraction compositions of the compounds CH_4 , CO , H_2 , and CO_2 of Figure 15 recalculated on dry basis. The symbols represent experimental data of a catalytic glucose steam gasification reported in Reference 39 for a reaction time of 30 s. The experiments are performed in a fluidized bed gasifier with residence times of 5, 10, 20, and 30 s. It is reported in this reference that at shorter reaction times the catalytic gasification is kinetically controlled and that the larger the reaction time the closer the experimental product distribution to the equilibrium distribution: the composition of hydrogen and carbon dioxide getting higher and those of carbon monoxide and methane lower. Even if the equilibrium limit compositions are not reached in these experiments, the trends of their evolution for larger residence times are well predicted by the thermodynamic equilibrium models.

3.3 | Steam gasification of cellulose

Cellulose, the most abundant organic polymer, is a structural component of cell wall in green plants. It is a linear polysaccharide with the chemical formula $[\text{C}_6\text{H}_{10}\text{O}_5]_n$ with n varying from hundreds to thousands.

The overall reaction between cellulose and steam can be expressed by the transformation

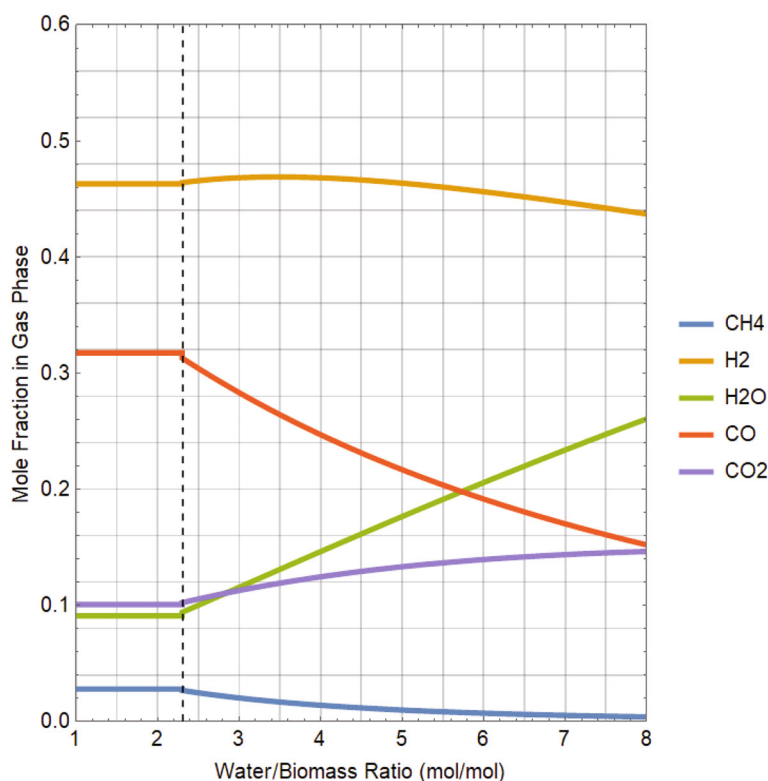
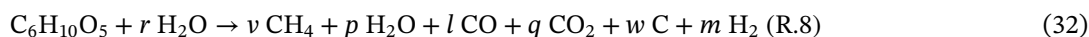


FIGURE 15 Effect of water/glucose-ratio on the gas phase composition at $P = 1$ bar and $T = 973.15$ K (700°C)

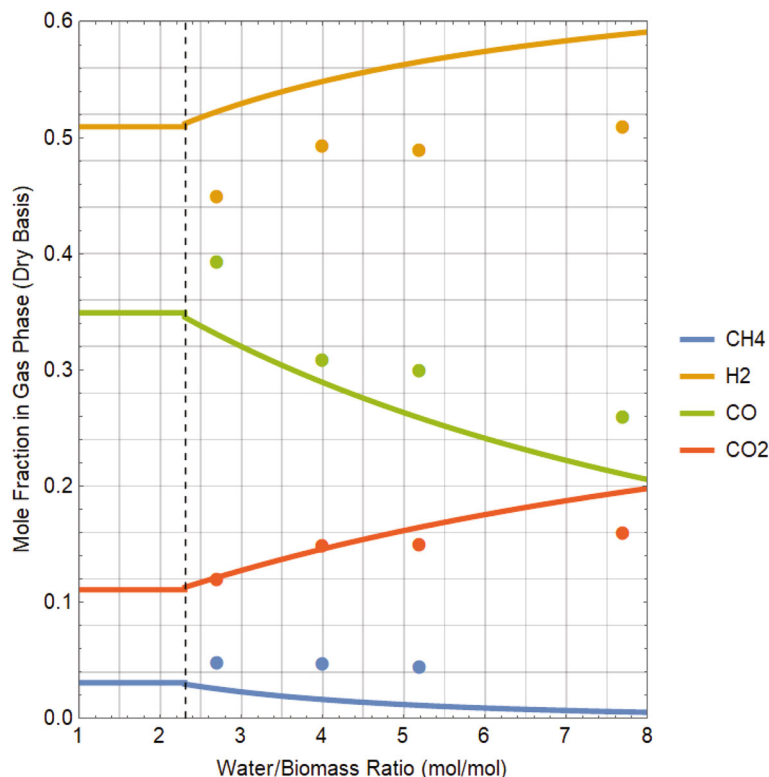


FIGURE 16 Effect of water/glucose-ratio on the gas phase composition (dry basis) at $P = 1$ bar and $T = 973.15$ K (700°C) in comparison with experimental data for a residence time of 30 s³⁹

where

$$q = \left(\frac{5 + r - l - p}{2} \right); v = \left(\frac{5 + r - m - p}{2} \right); w = \left(1 + p - r + \frac{m - l}{2} \right) \quad (33)$$

The equality constraints are easily derived

$$\text{C} : n_{\text{CH}_4} + n_{\text{CO}} + n_{\text{CO}_2} + n_{\text{C}} = 6 \quad (34)$$

$$\text{H} : 4n_{\text{CH}_4} + 2n_{\text{H}_2} + 2n_{\text{H}_2\text{O}} = 10 + 2r \quad (35)$$

$$\text{O} : n_{\text{H}_2\text{O}} + n_{\text{CO}} + 2n_{\text{CO}_2} = 5 + r \quad (36)$$

For the elementary steps occurring in the gasification process of cellulose using steam, Table 9 still applies.^{37,38}

3.3.1 | Effect of pressure and temperature

The effects of pressure and temperature on the gasification equilibrium of cellulose with steam are similar to those already discussed in the preceding case. Figure 17 shows the equilibrium composition versus pressure when the initial cellulose to steam molar ratio is equal to unity and $T = 1000$ K. The solid lines represent the gas-phase water-free mole fractions for CH_4 , H_2 , CO , and CO_2 obtained by our calculation using MATHEMATICA® while the colored squares are their counterparts gathered from RGIBBS of ASPEN-Plus®. Clearly, the match between the two sets of data is very good.

Figure 18 provides the equilibrium composition versus temperature when the initial cellulose to steam molar ratio is equal to unity and $P = 1$ bar. The solid lines represent the overall mole fractions of the different components obtained by our calculation using MATHEMATICA® while the colored dots are their counterparts gathered from digitizing fig. 1 of

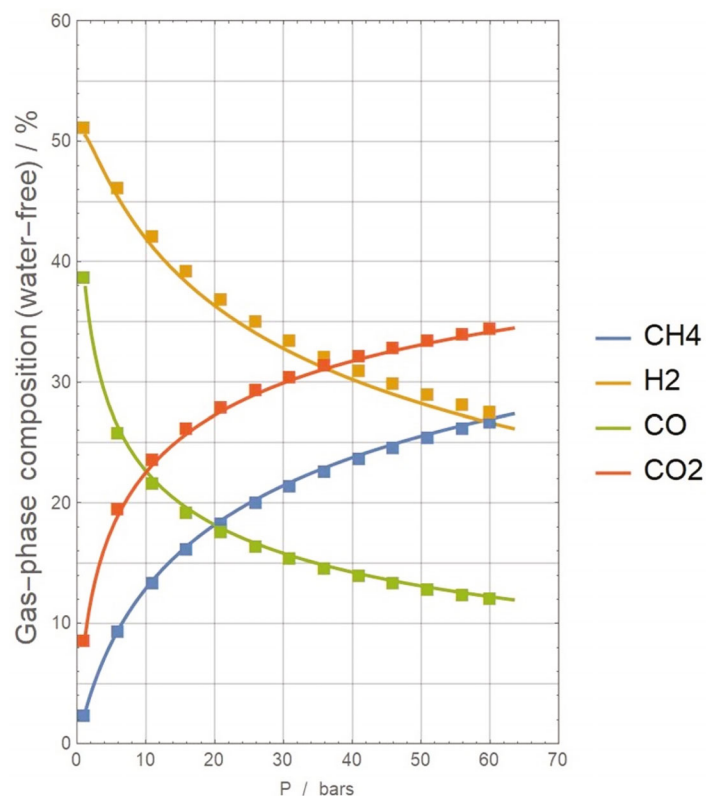


FIGURE 17 Equilibrium curves for a gas containing 1 mole of cellulose and 1 mole of H_2O at $T = 1000$ K (solid lines: present calculation and squares: data computed by RGIBBS of ASPEN-Plus®)

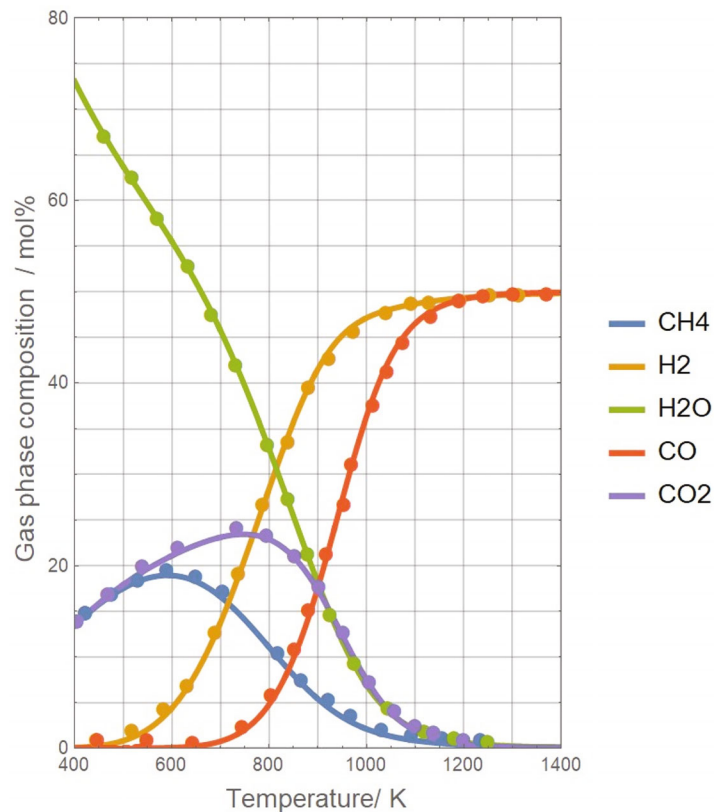


FIGURE 18 Equilibrium gas-phase composition versus temperature at $P = 1$ bar for an initial mixture containing 1 mole of water for 1 mole of cellulose (solid lines: present calculation; dots: data from Reference 40)

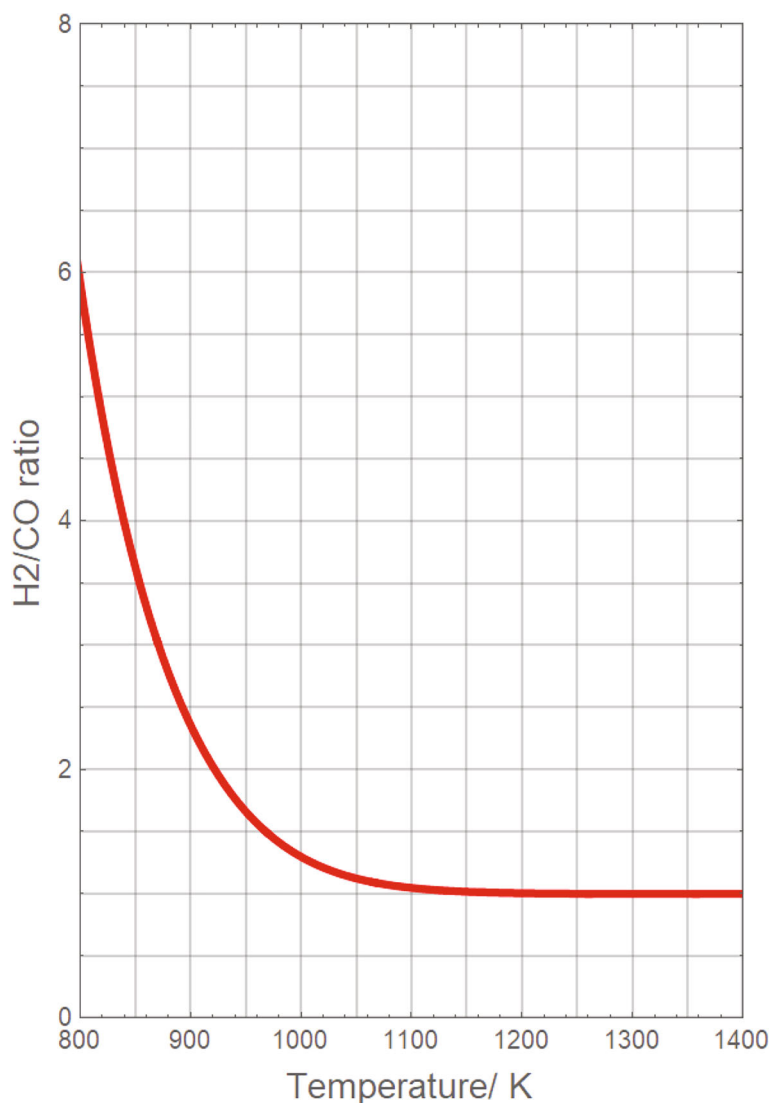


FIGURE 19 H_2/CO ratio versus T for an initial mixture containing 1 mole of cellulose and 1 mole of H_2O at $P = 1$ bar

Reference 40. Again, the match between the two sets of data is excellent. We notice further that high temperatures and low pressures favor the syngas production by the cellulose gasification process, similar to that of glucose gasification process.

Figure 19 depicts the ratio H_2/CO versus T for an initial cellulose to steam molar ratio equals to unity and $P = 1$ bar. This ratio decreases with increasing temperature until it reaches unity.

It should here again noticed that calculations for temperature and pressure effects have been performed using both equilibrium approaches (S and NS) with identical results.

4 | CONCLUSION

In the present paper, we compare the two routes generally adopted in order to predict the equilibrium composition of complex reacting systems. These two approaches are: (1) the stoichiometric method, which uses the equilibrium constants of a selected set of independent chemical reactions relating reactants and products of the chemical system and (2) the non-stoichiometric method where one minimizes the total Gibbs free energy of the assumed set of chemical species. Moreover, we point out the condition, derived from linear algebra, which must be satisfied in order for both methods to yield the same equilibrium compositions. We illustrate the above points by studying the high pressure and high temperature combustion of hydrazine and propane as well as the gasification of glucose and cellulose. We show that the

stoichiometric approach yield the same solution as the non-stoichiometric method for all the reacting system considered in the present paper.

Once a specific approach has been adopted, the mathematical problem at hand can be solved using one of the various numerical techniques available to us (e.g., minimization or root-finding methods). In particular, for the hydrazine and propane combustions, we test several numerical techniques and show that they yield essentially identical results.

Arc-length continuation represents a powerful numerical technique suitable for parametric studies. To our knowledge, this technique has never been used in order to investigate the effect of the relevant parameters (e.g., pressure, temperature, and feed composition) on the equilibrium composition of complex reacting systems. Hence, we develop an algorithm in order to implement the arc-length continuation technique and to study chemical reaction equilibria. The resulting equilibrium composition of three selected case studies (i.e., ammonia synthesis and gasification of glucose and cellulose) are successfully benchmarked against simulation data based on the RGIBBS unit of the process simulator: Aspen-Plus®. Further, we found that for the few cases where experimental data are available in the open literature the classical thermodynamic equilibria with relatively simple equations of state (ideal gas law, Peng-Robinson equation of state) does effectively predict real world data.

Final remark: All computations were performed using a single technical computing environment: MATHEMATICA®.

PEER REVIEW

The peer review history for this article is available at <https://publons.com/publon/10.1002/eng2.12506>.

DATA AVAILABILITY STATEMENT

The computer notebooks are available upon request from the corresponding author.

CONFLICT OF INTEREST

Authors have no conflict of interest relevant to this article.

ORCID

Housam Binous  <https://orcid.org/0000-0002-3196-1870>

Ahmed Bellagi  <https://orcid.org/0000-0001-6934-2007>

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APPENDIX A. PENG-ROBINSON EQUATION OF STATE

The Peng-Robinson Equation of State (PR EoS) was developed primarily for hydrocarbons but can also handle other common compounds of interest such as water, carbon dioxide and monoxide, nitrogen, hydrogen, and oxygen.^{41,42} An additional advantage of such real EoS resides in its ability to describe both liquid-phase and vapor-phase behavior. The PR EoS relates molar volume v to pressure P and temperature T and writes^{41,42}:

$$P = \frac{R T}{v - b} - \frac{a \alpha(T)}{v(v + b) + b(v - b)} \quad (\text{A1})$$

where

$$a = 0.45724 \frac{R^2 T_c^2}{P_c}, b = 0.0778 \frac{R T_c}{P_c} \quad (\text{A2})$$

$$\alpha(T) = \left[1 + (0.37464 + 1.54226 \omega - 0.26992 \omega^2) (1 - \sqrt{T_r}) \right]^2 \quad (\text{A3})$$

and

$$T_r = \frac{T}{T_c} \quad (\text{A4})$$

The PR EoS can be extended to mixtures using the mixing and combining rules given below^{41,43}:

$$P = \frac{RT}{v_M - b_M} - \frac{a_M}{v_M (v_M + b_M) + b_M (v_M - b_M)} \quad (\text{A5})$$

With mixture a_M and b_M related to their individual components counterparts by:

$$b_M = \sum_{i=1}^n y_i b_i; a_M = \sum_{i=1}^n \sum_{j=1}^n y_i y_j a_{ij}; a_{ij} = \sqrt{a_i a_j} (1 - k_{ij}) \quad (\text{A6})$$

$$a_i = \frac{0.45724 \alpha_i(T) R^2 T_{ci}^2}{P_{ci}}; b_i = \frac{0.0778 R T_{ci}}{P_{ci}} \quad (\text{A7})$$

$$\alpha_i(T) = \left[1 + \beta_i (1 - \sqrt{T_{ri}}) \right]^2$$

$$\text{with } \beta_i = 0.37464 + 1.54226 \omega_i - 0.26992 \omega_i^2 \quad (\text{A8})$$

and

$$T_{ri} = \frac{T}{T_{ci}} \quad (\text{A9})$$

The binary interaction parameters, k_{ij} , are usually set equal to zero if unavailable.^{41,42}

The residual molar Gibbs free energy for pure component is given by^{43,44}:

$$\frac{G^R}{R T} = Z - 1 - \ln \left(Z - \frac{P b}{R T} \right) - \frac{a}{2\sqrt{2} b R T} \ln \left(\frac{v + (1 + \sqrt{2}) b}{v + (1 - \sqrt{2}) b} \right) \quad (\text{A10})$$

where $Z = Pv/RT$ is the compressibility factor. For a mixture, the residual contribution G_M^R is then

$$\frac{G_M^R}{R T} = Z - 1 - \ln \left(Z - \frac{P b_M}{R T} \right) - \frac{a_M}{2\sqrt{2} b_M R T} \ln \left(\frac{v_M + (1 + \sqrt{2}) b_M}{v_M + (1 - \sqrt{2}) b_M} \right) \quad (\text{A11})$$

with $Z = Pv_M/RT$. The mixture molar volume v_M is determined from the roots of the cubic PR EoS. At a specific temperature and a pressure, up to three values for Z or v_M can be obtained. If two roots of the cubic equation, which has real coefficients, are complex conjugate then there is a single phase in the system (i.e., either a gas or a liquid phase). On the other hand, if all three values are real then we have a two-phase system. In such case, the largest value corresponds to the vapor-phase property (i.e., Z or v_M) while the smallest one is its liquid-phase counterpart.⁴⁵

APPENDIX B. ARC-LENGTH CONTINUATION TECHNIQUE

B.1 General considerations

Let us consider the following parametric system of N nonlinear equations in N unknowns $(m_1, m_2, \dots, m_N)$, written in vector representation

$$\mathbf{F}(\mathbf{m}; \varphi) = 0 \quad (\text{B1})$$

φ is a parameter on which the solution depends. The components of vectors $\mathbf{F}$ and $\mathbf{m}$ are respectively, $\mathbf{F} = (f_1, f_2, \dots, f_N)$ and $\mathbf{m} = (m_1, m_2, \dots, m_N)$. The objective is now to obtain a solution of such a system as a continuous function of φ . In the arc-length continuation method, unknown variables and parameter φ are set as a function of the arc-length,^{7,8} represented by s in the following.

Let us now admit that for a particular value φ_0 of the parameter φ , the equations set (B1) has a solution $\mathbf{m}_0$,^{46–48}

$$\mathbf{F}(\mathbf{m}_0; \varphi_0) = 0 \quad (\text{B2})$$

and that the solution $\mathbf{m}$ is an analytic function of φ (i.e., continuous and differentiable). In this case, a neighboring solution $\mathbf{m} = \mathbf{m}(\varphi)$ can be calculated by constructing a Taylor expansion about φ_0 . For a multi-value function of φ , the solution can be parameterized in term of arc-length s on the solution curve. Therefore, we can write:

$$\mathbf{m} = \mathbf{m}(s), \varphi = \varphi(s) \quad (\text{B3})$$

So that Equations (B1) become

$$\mathbf{F}(\mathbf{m}(s), \varphi(s)) = 0 \quad (\text{B4})$$

To solve Equation (B4), one more relation is needed. The appropriate auxiliary equation is given by the definition of the arc-length:

$$\frac{d\mathbf{m}}{ds} \cdot \frac{d\mathbf{m}}{ds} + \left(\frac{d\varphi}{ds} \right)^2 = 1 \quad (\text{B5})$$

The system to solve (B4) and (B5) is now a differential algebraic equations (DAE) set in $(N + 1)$ unknowns. Although, one can solve this DAE using the built-in MATHEMATICA® function `NDSolve`,⁴⁶ we choose in the present article to discretize Equation (B5).^{46–48} The resulting algebraic system is then solved with the built-in MATHEMATICA® function `FindRoot` in combination with the continuation procedure `NestList`. This latter MATHEMATICA® function is very handy when one wishes to make an iterative computation using every previously found solution as a starting point for the next iteration or equilibrium calculation.

APPENDIX C. GIBBS FREE ENERGY MINIMIZATION

In order to estimate the equilibrium composition at fixed temperature and pressure of a complex chemical system composed of several known components and subject (or not) to a certain number of constraints, one method is the minimization of the mixture Gibbs free energy G_M ,^{1,2,6}

$$\min G_M(n_i) \quad (\text{C1})$$

Since this state function depends on the number of moles n_i of each of the involved chemical species i , these latter quantities can be taken as decision variables. This is a constrained minimization problem: The set of n_i -values that minimize G_M must verify equality constraints expressing the conservation of the atomic elements. Denoting with α_{ki} the number of element k in the chemical formula of component i and with ζ_k the total number of moles of element k initially present in the feed, these equality constraints write

$$\sum_i n_i \alpha_{ik} = \zeta_k \quad (C2)$$

Another constraint must be considered when a real fluid equation of state is used to describe the volumetric behavior of the equilibrium mixture. In the case of the PR EoS, the constraint writes

$$\mathcal{F}_Z = Z^3 + (b_M - 1) Z^2 + (a_M - 3 b_M^2 - 2 b_M) Z - a_M b_M + b_M^2 + b_M^3 = 0 \quad (C3)$$

Because we are dealing with only equality constraints, the optimum solution can be obtained by introducing the Lagrangian multipliers, noted by λ_k for the w chemical elements, and λ_Z for the compressibility factor. The problem at hand consists now in minimizing the Lagrangian function $\mathcal{L}$ defined as:

$$\mathcal{L} = G_M(n_i) + \sum_{k=1..w} \lambda_k \left[\left(\sum_{i=1..N} n_i \alpha_{ik} \right) - \zeta_k \right] + \lambda_Z \mathcal{F}_Z \quad (C4)$$

The minimization problem is thus converted into the following system of $(w + N + 2)$ nonlinear equations⁴⁹:

$$\begin{cases} \left(\frac{\partial \mathcal{L}}{\partial \lambda_k} \right)_{n_i, Z, \lambda_{j \neq k}, \lambda_Z} = 0 \quad (k = 1, 2, 3, \dots, w) \\ \left(\frac{\partial \mathcal{L}}{\partial \lambda_Z} \right)_{n_i, Z, \lambda_k} = 0 \\ \left(\frac{\partial \mathcal{L}}{\partial n_i} \right)_{n_{k \neq i}, Z, \lambda_k, \lambda_Z} = 0 \quad (i = 1, 2, 3, \dots, N) \\ \left(\frac{\partial \mathcal{L}}{\partial Z} \right)_{n_i, \lambda_k, \lambda_Z} = 0 \end{cases} \quad (C5)$$

The unknowns in the above system are the N number of moles present in the system (i.e., n_i for $i = 1, 2, 3, \dots, N$), the compressibility factor (i.e., Z) and $(w + 1)$ Lagrangian multipliers (i.e., λ_k for $k = 1, 2, 3, \dots, w$ and λ_Z).

In order to compute the Gibbs free energy for a multi-phase system, the Gibbs free energy of all phases are added.⁴⁵ We can subdivide the contributions into two main parts:

- i Gibbs free energy for the gas and liquid phases. In fact, the calculation of both of these terms involves applying the PR EoS and taking either the liquid or gas phase molar volume (or equivalently the liquid or gas phase compression factor).
- ii Gibbs free energy of the solid phase.

C.1. Gas and liquid phase mixture Gibbs free energy

For the gas-phase, the free energy of the mixture, $G_{g,M}$, is the sum of the ideal gas contributions, $G_{g,M}^o$, and the residual contributions, G_M^R ⁴¹:

$$G_{g,M} = G_{g,M}^o + G_M^R \quad (C6)$$

$G_{g,M}^o$ is given by:

$$G_{g,M}^o = n \sum_{i=1}^n y_i (g_i^o + R T \ln y_i) \quad (C7)$$

In the above equation, the pure component molar ideal gas Gibbs free energy or g_i^o is expressed by:

$$g_i^o = g_i^o(T_0) + \int_{T_0}^T C_{p,i} dT - T \int_{T_0}^T \frac{C_{p,i}}{T} dT - S_i(T_0) (T - T_0) + R T \ln \left(\frac{P}{P^0} \right) \quad (C8)$$

It is usual to take $T_0 = 298.15$ K and $P^0 = 1$ bar as reference temperature and pressure, respectively.

Although it is possible to use the above expression in order to calculate g_i^o , we chose a different route in the present paper. Indeed, we fit the experimental values of g_i^o versus T at $P^0 = 1$ bar. Such values are available in several sources of thermochemical data of pure substances (e.g., Reference 13). For the sensitivity analysis of the equilibria, analytical expressions for $g_i^o(T, P^0)$ are needed, so we select the following empirical expression for $g_i^o(T, P^0)$:

$$g_i^o(T, P^0) = a_0 + a_1 T + a_2 T^2 + a_3 T^3 + a_4 T^4 + a_5 T^5 + a_6 T^6 + a_7 T^7 + a_8 T^8 + a_9 T \ln(T) \quad (C9)$$

Finally, we include the pressure-dependency:

$$g_i^o(T, P) = g_i^o(T, P^0) + R T \ln(P/P^0) \quad (C10)$$

It should be noted that the parameters $\{a_i, i = 1..9\}$ are determined separately for every pure component of the chemical system under study using the same platform of computation (i.e., FindFit of MATHEMATICA®).

C.2. Solid phase Gibbs free energy

Because solid–solid interactions are generally negligible, the total Gibbs free energy of a solid phase is merely the sum of all contributions from each component existing in the solid phase considered as present alone. Hence, the equation below applies:

$$n_s g_{s,M} = \sum_j n_{j,s} g_{j,s} \quad (C11)$$

In addition, the Gibbs free energy of any solid species (e.g., carbon in the case of gasification of biomass) is taken as a function of temperature only. In fact, this solid phase property is not strongly affected by pressure. Here, we make use again of the experimental thermochemical data available in Reference 13 to generate an interpolating function using the built-in command of MATHEMATICA® called Interpolation.

Finally, the expression of the total Gibbs free energy of a system with gas and solid phases writes as follows:

$$G^{\text{total}} = n_g g_{g,M} + n_s g_s \quad (C12)$$

C.3. Application of arc-length continuation to reacting systems

C.3.1. Temperature or pressure parametric study

In order to study the effect of temperature, we fix the pressure (e.g., $P = P_0$) and consider temperature as a parameter. In the system of nonlinear equations Equation (B1)

$$\mathbf{F}(\mathbf{m}; \varphi) = 0$$

φ is now the temperature T , and the unknowns vector of dimension $(N + w + 2)$ $\mathbf{m} = (n_1, n_2, \dots, n_N, Z, \lambda_1, \lambda_2, \dots, \lambda_w, \lambda_Z)$.

The effect of pressure is studied using a similar approach. This time, we fix the temperature (e.g., $T = T_0$) and consider pressure as a parameter (i.e., we set $\varphi = P$).

C.3.2. Feed composition parametric study

In order to study the effect of the initial composition of the reacting system, we fix both the pressure and temperature (e.g., $P = P_0$ and $T = T_0$) and consider the initial number of moles of one of the components of the system, ϑ , as a parameter. Hence, in the system of nonlinear equations, $\mathbf{F}(\mathbf{m}; \varphi) = 0$, $\varphi = \vartheta$. This latter step requires an appropriate adjustment of the parameter: A_k . Again, we have obtained the DAE to be solved using FindRoot in conjunction with NestList as indicated previously.

C.3.3. Solution procedure for equilibrium calculation according to the NS-method

We give in the following the MATHEMATICA® approach applied when the effect of temperature on the glucose (or cellulose) gasification equilibrium is studied.

- Calculate the total Gibbs free energy for the fixed pressure.
 1. Declare SRK EoS interaction parameters for gasification products.
 2. Choose the operating pressure, P .
 3. Define the ideal gas contribution to the Gibbs free energy using Equation (C8), the cubic equation for the compressibility factor Z using Equation (C3), the residual Gibbs free energy using Equation (A11) and the Gibbs free energy for solids (i.e., coke).
Sum all contributions to get the overall Gibbs free energy.
- Define the Lagrangian function based on the previously defined total Gibbs free energy and the different equality constraints (the elemental material balance equations for C, N, H, and O and the cubic equation for Z). Here, one has to choose the feed stream composition or equivalently the glucose to steam molar ratio, r .
- Set the partial derivatives with respect to the decision variables (i.e., the components' mole numbers n_i and Z) and the Lagrangian multipliers, λ_i .

Equal to zero to get a system of algebraic equations labeled SYS.

- Add the auxiliary equation the usual differential equation labeled AUX-DIFF⁸ to be used with **NDSolve** (Case 1) or alternatively the discretized version labeled AUX-DISC⁷ to be used with **NestList** (Case 2).

Case 1

- Set all mole numbers, Z , T and Lagrangian multipliers, λ_i , as functions of the arc-length parameter, s .
- Compute an initial solution by solving the system of algebraic equations SYS for one specific value of T (e.g., $T(s=0) = 700\text{K}$) with **FindRoot**.
- Use **NDSolve** to integrate the system of DAEs (SYS + AUX-DIFF) with the initial condition found previously.

Case 2

- Compute two initial solution by solving the system of algebraic equations SYS for two close values of T (e.g., $T = 700\text{K}$ and 700.1K) with **FindRoot**.
- Use **NestList** to solve iteratively the augmented system of algebraic equations (SYS+AUX-DISC) for a large range of increasing temperature values.